

Design, Testing, and Reception of Conversational AI Agents in Financial Services

Master Thesis



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By
Felipe Miguel Wallner Giacomelli

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Felipe Miguel Wallner Giacomelli - s244086

.....
Signature

.....
Date

*A minha tia Sara,
por seu amor e suporte
durante minha estada no exterior.*

*To my aunt Sara,
for her love and support
during my stay abroad.*

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I would like to thank my supervisor, Lars Alkærsig, for his guidance on the market feasibility study and the lean startup methodology underpinning this work.

*Dicen que soy pobre
Y no sé cuál es mi pobreza
Si me la agarro con las dos manos
Y me sobra la cabeza.*

Abstract

This thesis reports on the design, testing, and reception of a conversational AI agent built to perform sales-lead qualification in Brazilian financial services, and on the commercial feasibility of such an agent as a product. The agent was deployed on WhatsApp at a host fintech and conducted interactions in Brazilian Portuguese against the firm’s qualification heuristic. It is treated throughout as a workflow-based sociotechnical system rather than as a single language-model prompt. The work is organised as a single case study with a triangulated evidence base of workflow artefacts, pilot conversations, post-deployment interviews with ideal-customer-profile (ICP) participants, and the researcher’s experience as a human SDR within the host company. The artefact was developed across Build-Measure-Learn cycles and evaluated against a testing framework of six dimensions, a customer simulator, and a simulation harness combining LLM-as-judge scoring with deterministic checks. Three design insights emerged. Domain grounding and workflow structure carried the artefact’s behaviour more visibly than backbone model choice, as evidenced by a controlled cross-vendor substitution that preserved qualification behaviour while shifting conversational register. Explicit stopping logic was necessary for acceptable operational behaviour. An attempt at outbound cold-list automation on WhatsApp was not viable in this context, and the work pivoted to inbound deployment as a substantive learning outcome. Reception was assessed through a small set of post-deployment interviews with ICP participants who interacted with the agent in a guided role-play scenario. The evidence is treated as exploratory validation rather than as a standalone reception study, as no large-N or controlled-exposure design was conducted. It suggests that trust in AI for financial-services interactions is uneven across participants. The feasibility study finds that the addressable need exists in principle, but that commercial viability depends materially on acquisition strategy and channel access rather than on agent quality alone. Findings are interpreted in light of the single-case design and the lead-engagement gap between the human SDR’s inputs and the agent’s.

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Chapter 1

Introduction

1.1 Problem and motivation

With the digitalisation of financial services, businesses increasingly acquire potential customers through digital marketing rather than through traditional broker networks or physical branches. Advertising campaigns are designed to attract users matching a firm’s *ideal customer profile* (ICP), and prospective customers who interact with these campaigns typically register their contact details through a landing page or online form. Each such registration produces what is termed a *lead*.

Not every registration represents a viable commercial opportunity. The proportion of leads with genuine purchasing intent is small, and a firm must filter the inbound flow before its sales team invests time in conversation. This filtering process is known as *lead qualification*.

Lead qualification is particularly consequential in financial services. Eligibility for the product is constrained by the customer’s financial position as much as by their stated interest: a prospective buyer of a credit product, for instance, must hold sufficient collateral and be able to service the obligation. Stated willingness alone is not sufficient.

In principle, the qualification task can be performed by the same salesmen who close the eventual sale. In practice, qualification and closing are distinct activities. Qualification is a repetitive procedure in which a small, pre-defined set of facts must be established about each lead, while closing is an interpersonal negotiation requiring product expertise and judgement.

The hourly cost of a salesman capable of closing a financial-services transaction is high, and the marginal value of that salesman’s time is greatest when applied to leads that have already cleared qualification. For this reason, firms commonly assign qualification to a dedicated role, the *Sales Development Representative* (SDR), whose function is to determine which leads warrant the sales team’s attention.

The qualification task has properties that suggest it is amenable to automation. It is repetitive, follows a roughly fixed conversational script, requires the extraction of a small set of structured facts, and is bounded by criteria the firm can specify in advance. Recent advances in Artificial Intelligence (AI), in particular large language models capable of multi-turn dialogue and tool use, raise whether the SDR role, or part of it, can be performed by an automated agent. This thesis investigates that question.

1.2 Thesis structure

The remainder of this thesis is organised in six chapters. In **Chap. 2**, the relevant literature on conversational AI agents, sales development work in financial services, the WhatsApp channel, and user trust in AI is reviewed. The research design adopted in this thesis, including its central research objective and sub-questions, is then presented.

In **Chap. 3**, the case context is introduced. The business setting and ICP of the host firm, the qualification task and its lead-status model, the conversation design principles followed by the agent, and its system architecture are described.

In **Chap. 4**, the testing framework developed to evaluate the agent during iterative development is presented. Its evaluation dimensions, test-case design, simulation harness, and limitations are discussed.

In **Chap. 5**, the artefact’s iterative development is reported through the Build-Measure-Learn cycles executed during the project. Each cycle describes what was built, what was measured, and what was learned. The chapter closes with a post-deployment exploratory study with participants matching the ICP.

In **Chap. 6**, the commercial feasibility of the agent as a product is assessed. The unmet need, the artefact’s fit to that need, the willingness of Brazilian financial-services firms to pay for such an agent, and the size of the addressable market are examined.

Finally, in **Chap. 7**, the findings are discussed against the research questions. Then, the limitations of the work are stated, and the contributions and directions for future research are presented.

Chapter 2

Background, Related Work, and Research Design

This chapter establishes the foundation of the thesis. It begins with four sections of literature review, covering conversational AI agents, sales development representative work in financial services, WhatsApp as a business communication channel, and user trust and bias toward AI agents. The chapter concludes with the research design under which the artefact was built and evaluated. The empirical work is organised as a single-case study, an in-depth investigation of one situated artefact in its real-world setting, supported by an evidence base that combines quantitative and qualitative sources to triangulate findings.

2.1 Conversational AI agents

Conversational AI agents are software systems that interact with users through multi-turn natural-language dialogue [1]. The term spans a wide design space, from open-domain assistants such as commercial chatbots [2] to narrow task-oriented systems that pursue specific goals [3]. The present thesis is concerned with the second class: agents operating within a bounded domain and expected to drive the conversation toward a determined outcome.

The recent generation of conversational agents has been enabled by large language models (LLMs), built on the transformer architecture [4] — exemplified by the GPT family of models [5]. LLMs supply the linguistic competence required to handle free-form user input, generate fluent responses, and follow task-relevant instructions through prompting rather than through explicitly programmed rules.

Task-oriented dialogue is a mature area of research within conversational AI, concerned with systems that pursue user goals such as booking a flight, completing a form, or resolving a query. Recent surveys document the rapid evolution of the field after the introduction of LLMs, with comprehensive reviews covering both open-domain and task-oriented systems based on LLMs [1].

A persistent challenge in this area is the maintenance of state across many turns. The agent must track what the user has said, what is still required to complete the task, and what external information is needed. Recent evaluations of LLM-based dialogue systems show that coherence across many turns and grounded task completion remain difficult even for capable models [3, 1].

This challenge is most acute for agents operating over long, complex turn sequences. In the

SDR setting at hand, each user message is typically brief, and the conversational history can be persisted to a database. Therefore, the difficulty is comparatively mild.

The LLM need not handle every operation an agent performs. Tasks that are inherently deterministic, such as record lookups, status checks, and persistence of conversation state, are more reliably handled by explicit code than by model inference. This division is exemplified in recent work on task-oriented dialogue [6]: the LLM handles language understanding and generation, while explicit code governs business logic. This separation allows reliable task execution to coexist with the linguistic flexibility of an LLM.

A complementary line of work concerns tool use, situated within the broader trend toward LLM-based autonomous agents that operate beyond a single conversation [7]. Tool use refers to the agent’s ability to invoke an external function, exposed through a declared interface, and to incorporate its return value into subsequent generation. Recent work shows that this capability can be acquired by the LLM itself with little explicit supervision: the model learns both when an external call is warranted and how to fold the returned information back into its output [8]. Tool invocation extends the agent’s reach outside the dialogue itself, enabling access to large catalogues of external APIs [9]. Beyond per-turn invocation, recent work converts sequences of LLM tool reasoning into reusable workflows, allowing the agent to call a procedure validated in advance rather than derive one again each time [10].

The artefact studied in the present thesis is an instance of this hybrid LLM and workflow architecture. Its language understanding and generation are delegated to an LLM, while a deterministic workflow governs state and stop conditions. Tool calls extend the agent’s capability beyond the conversation. The artefact’s specific design is presented in [Sec. 3.4](#).

2.2 SDR work and financial-services communication

Sales development bridges marketing and sales. Leads enter through marketing campaigns or self-initiated contact, and the sales development representative qualifies them by intent and capability before passing them to a salesman. The task is conversational by nature, and it has long been a target for automation because qualification decisions follow recognisable patterns.

Recent work in information systems examines how customers respond to automated sales agents and how those responses shift across sales stages [11]. The finding is that automated and human agents are received differently depending on where in the funnel the conversation occurs. This has direct implications for any comparison between the two: a comparison performed at one stage cannot be transferred to another.

A complementary line of work studies how chatbot message personalisation shapes consumer purchase intention [12]. Personalised chatbot messages were reported to produce higher purchase intention than non-personalised ones, with the effect mediated by the consumer’s perceived control over the shopping experience.

Other controlled experiments document how specific design elements of the chatbot affect downstream consumer outcomes. A randomised study of welcome messages reports that a humorous greeting increases both engagement and conversion-to-lead relative to a neutral greeting [13]. In addition, a complementary work shows that the presence of a chatbot on a brand website shifts the perceived value of products and increases purchase intention [14].

A separate strand of evidence concerns how human-like design choices shape consumer perception. A series of experiments on conversational commerce compared chatbots with and without anthropomorphic verbal cues such as a human name, first-person pronouns,

and small talk. Chatbots with these cues were reported to produce higher perceived personalisation than those framed as generic automated systems [15].

Financial-services communication imposes constraints that distinguish it from general sales work. A 2023 report by the United States Consumer Financial Protection Bureau finds that chatbot adoption is universal across the ten largest US commercial banks. However, chatbots' performance degrades as the consumer's query becomes more complex, and that poorly designed deployments can damage consumer confidence and expose institutions to legal liability [16]. The constraints are regulatory, technical, and reputational.

These constraints are not specific to the US market. A 2025 industry survey of Brazilian credit fintechs reports that 67% of the respondents are currently developing or studying AI-based solutions, yet generative AI in customer-facing roles is approached with caution [17]. In the absence of AI-specific regulation, the sector prioritises data security and compliance with Brazil's general data-protection law (LGPD). The Brazilian fintech setting therefore mirrors the regulatory and reputational pressures of the US banking case.

The thesis artefact occupies the intersection of the chatbot-in-sales literature and the constraints of financial-services communication. Studies of LLM-based agents operating in regulated, non-English, mobile-instant-messaging sales contexts remain scarce.

2.3 WhatsApp as a business channel

Mobile instant messaging (MIM) applications are increasingly used as a business communication channel, replacing or supplementing email, SMS, and voice in customer communication. Their appeal is asynchronous one-to-one dialogue at low marginal cost, on a device the user already carries, with conversation history preserved across sessions.

In Brazil, WhatsApp occupies a particularly dominant position. The country is WhatsApp's third-largest market globally and its largest market outside Asia, with industry data reporting active use by 93% of the Brazilian population and an estimated 148 million users in 2024 [18]. The app ranks first by usage frequency among social platforms in the country.

Despite this penetration, end-user attitudes toward MIM as a business channel are not uniform. A segmentation study of 478 respondents identifies proactive, benevolent, and hesitant clusters, distinguished by effort expectancy, perceived usefulness, and ease of use of the channel for service interactions [19]. Ubiquity at the platform level does not translate into uniform receptivity at the user level.

For an SDR built on WhatsApp in Brazil, these two findings together define the conditions of operation. The platform makes the audience reachable in principle, but the segmentation pattern implies that some users will engage willingly with an automated message while others will not, regardless of conversational quality.

2.4 User trust and bias toward AI

How users come to trust or distrust AI agents bears directly on the design and reception of any conversational system intended to interact with them. The literature relevant to the present work spans three threads: how people perceive non-human agents, the documented patterns of human resistance to algorithmic systems, and the specific dynamics of AI adoption in financial services.

Anthropomorphism refers to the attribution of human characteristics, such as intentionality, emotional states, and personal traits, to non-human entities. People readily do so, and this tendency has been formalised as a three-factor model [20]. The model identifies the

activation of knowledge of human behaviour, the motivation to make sense of an agent’s behaviour, and the motivation to establish social connection as the conditions under which anthropomorphic perception is most likely to occur.

These tendencies underpin the *Computers Are Social Actors* paradigm, which holds that people apply social rules and expectations to interactive media as if they were human [21, 22]. The paradigm provides the scaffolding on which subsequent work on conversational-interface design has been built.

Social presence refers to the extent to which a medium allows users to experience another party as psychologically present [23]. Applied to conversational interfaces, anthropomorphic linguistic cues raise a chatbot’s perceived social presence. This mediates effects on a range of consumer outcomes including trust, willingness to purchase, recommendations to others, and overall satisfaction [24]. The mediation pattern indicates that the value of human-like cues lies in the social presence mechanism they activate rather than in any direct surface effect.

Anthropomorphism is therefore not only a perceptual phenomenon but a deliberate design lever. Two design choices have been examined experimentally: equipping the agent with a human-like visual identity (personification), and configuring its communication in a socially attentive register. Both raise the chatbot’s perceived social presence, which then drives downstream user judgments such as empathic perception, perceived trustworthiness, and post-interaction satisfaction [25]. The implication is that the impact of human-like cues operates through social presence, not through the cues taken in isolation.

Behavioural-outcome measures replicate this pattern. A controlled experiment finds that linguistic cues signalling humanness raise the rate at which users comply with requests issued by the chatbot, with social presence again acting as the mediator [26].

On the receiving side of the relationship, a documented pattern of human resistance to algorithmic decision-making has been reported [27]. Brief exposure to algorithmic errors was enough to reduce participants’ confidence in it sharply. Given a choice between the algorithm and a less accurate human forecaster, they preferred the human. Distrust of automated systems can therefore be triggered even when the system is, on the evidence shown, more accurate than the human alternative.

However, the picture is less settled than this single finding suggests. Algorithmic appreciation has also been observed, with people weighting algorithmic advice more heavily than identical human advice in some conditions [28]. The strength of aversion appears to depend on the task: subjective tasks attract more resistance than objective ones, and shifting consumers’ perception of a task toward the objective end can reduce aversion [29].

In a commercial context, the disclosure of an agent’s machine identity itself has been shown to affect customer behaviour. A randomised field study run on more than 6,200 customers who received an outbound sales call finds that prefacing the call with a disclosure that the caller was a chatbot cut conversion by roughly 80%, from a purchase rate of 23,7% to 4,8% [30]. The same chatbots, when not disclosed, performed comparably with experienced human agents [30]. Therefore, the bias under study is not only a matter of perceived competence but of identity disclosure.

The adoption of AI in financial services has its own friction. A multi-country survey of 765 prospective robo-advisor users, recruited in North America, the United Kingdom, and Portugal, finds that the user’s general attitude toward such systems, together with subjective norms transmitted through both mass media and personal contacts, drive

adoption decisions; prior familiarity with robotics moderates the strength of those drivers [31].

A more direct parallel to the present thesis examines the adoption of intelligent virtual assistants (IVAs) in financial services [32]. The study surveyed 377 fintech users after a chatbot interaction, modelling two user attitudes toward IVAs — hedonic (enjoyment) and utilitarian (perceived usefulness) — as predicted by two sets of constructs. The first set is AI-related, comprising perceived anthropomorphism, perceived intelligence, and perceived animacy. The second is sociopsychological, comprising technological self-efficacy and social presence. For financial-services adoption, the utilitarian dimension is the relevant one, since IVAs are not approached as a source of pleasure.

The three AI factors each had a significant effect on user attitudes. The need for human interaction acted as a positive moderator on the relationships between the AI factors and user attitudes.

Taken together, the literature establishes algorithm aversion as a documented phenomenon, and anthropomorphism as a deliberate design lever operating through social presence. AI adoption in financial services as conditioned by user characteristics that include familiarity with the technology and the need for human interaction.

The thesis artefact applies anthropomorphic cues as a literature-backed design principle [24, 25, 32], not as a hypothesis the thesis itself tests. The specific cues used by the agent are described in [Sec. 3.3](#). The reception evidence in this thesis is exploratory and is taken up in [Chap. 7](#).

2.5 Research design

Three choices structure the empirical work: a single-case study, a triangulated evidence base, and an iterative development process organised as Build-Measure-Learn cycles. The first two are introduced and justified in the present section. The third is named at the end of the section and expanded in [Chap. 5](#), where the cycles are enacted.

Within information systems research, Design Science Research (DSR) provides the canonical frame for studies that build and evaluate artefacts [33, 34]. The empirical work in this thesis is adjacent to DSR in that an artefact is built, observed iteratively, and reported on, but departs from it in emphasis. The contribution sought here is not a set of abstracted design principles, but a situated account of how a single artefact behaves and is received in its real-world setting. The case-study and Build-Measure-Learn framing adopted below was selected on those grounds.

Action research, a related tradition that engages the researcher as a participant in the system being studied, was also considered and not adopted. The empirical work here does not take the form of participatory cycles with multiple stakeholders co-producing the design, and the reception evidence is drawn from post-deployment interviews rather than from collaborative inquiry with the participants.

A case study is a research method that examines a phenomenon in depth, in its real-world setting, drawing on multiple sources of evidence [35]. It is the appropriate frame when context and phenomenon cannot be cleanly separated.

The AI SDR studied here cannot be cleanly separated from its organisational context. The lead sources, qualification goals, and communication channel are all shaped by the host company. Thus, the unit of analysis is a single sociotechnical system rather than a population of comparable systems.

A single case, rather than a comparative multi-case design, is appropriate for two reasons [35]. First, the research is exploratory in character: it addresses how an AI SDR can be designed and how it behaves under conditions that have not been systematically studied in the published literature. This favours theory building over theory testing [36]. Second, the case is revelatory [35]: it provides access to a phenomenon previously inaccessible to systematic study. The artefact was built to operate inside a regulated financial-services setting, on WhatsApp, in Brazilian Portuguese, with lead sources that the researcher could observe directly. A second comparable case would not have been accessible within the scope of this work.

The cost of this choice must be stated plainly. The findings reported in later chapters are grounded in one company and one domain, and no claim of broad generalisation is advanced. However, the design insights discussed in [Chap. 7](#) may serve as a starting point for financial-services firms offering comparable products.

The central research objective of this thesis is to investigate the design, testing, and reception of a conversational AI agent for lead qualification in Brazilian financial services, and to assess the commercial feasibility of such an agent as a product. The investigation is guided by the following main question:

How can a conversational AI agent for lead qualification in Brazilian financial services be designed and tested, and how is such an agent received by its intended users?

The main question is decomposed into four sub-questions:

1. *Design.* What design choices are required for an AI SDR to operate on WhatsApp in Brazilian financial services?
2. *Testing.* What evaluation dimensions and procedures are appropriate for assessing such an agent during iterative development?
3. *Reception.* How do ICP participants in Brazilian financial services report their experience with and attitudes toward AI SDR interactions in this domain?
4. *Feasibility.* Does a commercial opportunity exist for the agent as a product in Brazilian financial services, considering (a) whether there is an unmet need for automated lead qualification, (b) whether the artefact developed in this thesis meets that need, (c) whether Brazilian financial-services firms are willing to pay for such an agent, and (d) whether the market is large enough to sustain the business?

The scope of SQ3 warrants explicit statement. In this thesis, *reception* refers to qualitative, post-deployment evidence from ICP participants who interacted with the agent in a guided role-play scenario. It does not refer to a large-N reception study, nor to a controlled-exposure design in which participants are randomly assigned to AI and human interlocutors.

The thesis’s claim about reception is correspondingly limited: it concerns how a small set of ICP participants reported their experience, and is treated as exploratory validation rather than as a standalone reception study.

The empirical work draws on multiple evidence sources to address the four sub-questions. The sub-questions span design, testing, reception, and feasibility — no single evidence type answers all four. Convergent evidence from different sources is sought where possible, and divergent evidence is reported rather than reconciled [37].

Structural and behavioural evidence, namely the workflow artefact and pilot conversation

outcomes, supports claims about the system itself. Interpretive evidence, namely post-deployment interviews and qualitative review of conversation transcripts, supports claims about how users understood the agent.

The evidence base is enumerated here and developed in full in the chapters that follow:

1. *Workflow*. The AI SDR workflow itself, preserved as an *n8n* export and summarised in [Chap. 3](#).
2. *Pilot conversations*. The conversation logs produced by the deployed agent during pilot operation, taken up in [Chap. 5](#).
3. *Interviews*. A set of eleven post-deployment interviews with participants drawn from the ICP, taken up in [Chap. 5](#).
4. *SDR experience*. The researcher’s own experience as a human SDR within the host company, taken up in [Chap. 6](#).

In that role, the researcher handled engaged leads originating from the company’s own marketing campaigns, while the AI SDR handled leads drawn from a list that the company had purchased for outbound experiments.

This distinction is relevant for interpretation. The human and the agent did not face comparable inputs, and any comparison between the two is affected by the lead-engagement gap discussed in [Sec. 3.2.1](#) and by channel of origin. This gap is carried forward into [Chap. 6](#). The overall pilot evidence base remains small, and no strong sales-outcome claim is advanced on its basis.

A further feature of the design warrants disclosure. The researcher was both the designer of the artefact and the analyst of its behaviour and reception. This dual role is characteristic of design-oriented case studies in which the work is embedded in the setting it studies. However, it has interpretive consequences that are carried forward into the limitations discussion of [Chap. 7](#).

Neither the lead-engagement gap nor the researcher’s dual role can be fixed by adjusting the analysis. The lead-engagement gap is left in place: no causal claim is made from the human-vs-agent comparison, which is presented as a side-by-side description only.

The interpretive risk from the researcher’s dual role is reduced only through caution and through the disclosure above. The evaluation dimensions, the framing of conversational quality, and the interpretation of interview material were all chosen by the same person who built the system. An independent reviewer would likely emphasise different aspects. The evaluation framework of [Chap. 4](#) was developed inductively from pilot observations, so it does not act as an independent check.

The analytical approach maps to the evidence base. Conversations produced by the agent were reviewed against six evaluation dimensions introduced in [Chap. 4](#), which operationalise the behavioural and quality criteria the artefact was expected to satisfy. Interview material was subjected to thematic analysis, a method that identifies and reports patterns of meaning across qualitative data [38]. The coding was organised around recurring concerns raised by ICP participants, with the codes reported in [Chap. 7](#).

Workflow behaviour was analysed through direct inspection of the *n8n* export and through the audit log maintained by the lead-status management system.

The empirical work was not a single data-collection episode. It unfolded as a sequence of iterative cycles in which the artefact was built, its behaviour was observed in contact

with real leads, and the observations were used to revise the next increment. This iterative structure follows the Build-Measure-Learn logic of the Lean Startup tradition [39].

The Lean Startup framework is not introduced in the present chapter. Its apparatus is presented at the opening of [Chap. 5](#), where the cycles are enacted and the framework’s vocabulary is immediately applied. The apparatus comprises the definition of a cycle, the role of validated learning, and the definition of a pivot.

Chapter 3

Case Context and AI SDR Artefact

This chapter presents the case context and the AI SDR artefact. It opens with the business setting, the lead population the agent encounters, and the qualification task it performs. It closes with the conversation design principles that shaped the agent’s behaviour and the system architecture that realises them.

3.1 Business context and the ICP

The case is situated in financial services, a setting in which conversational AI deployment is established but not uncontested. In Brazil, a 2025 industry survey of credit fintechs reports that 67% of respondents are currently developing or studying AI-based solutions, although the use of generative AI in customer-facing roles is approached with caution [17]. The artefact studied here operates at the upstream end of this domain, in lead qualification. Here, the conversational task is bounded enough to be approached with a workflow-based agent.

The host company is a Brazilian financial-services firm. It offers credit and investment products to retail clients drawn from a high-net-worth segment. Two product lines are in scope for the present thesis: home equity credit and consortium.

In this context, home equity credit denotes a real-estate-secured loan in which the borrower pledges urban property as collateral in exchange for an interest rate substantially below that of unsecured retail credit. A consortium is a Brazilian collective-purchase scheme, regulated by the Central Bank of Brazil, in which a closed group of participants contributes monthly instalments to a common fund. Members successively draw access to a credit letter for a defined asset, by lottery or by competitive bidding, until the group is fully served.

The interaction channel is WhatsApp, accessed through Meta’s WhatsApp Business Cloud API. For a Brazilian retail-finance firm, the ubiquity of WhatsApp — discussed in [Sec. 2.3](#) — makes the platform the natural place for conversations between the firm and its customers. This is the platform for which the artefact was designed.

Financial services is a high-trust, high-friction commercial setting. Money is involved, prospective customers may be cautious or vulnerable, and poor communication carries reputational and compliance consequences that other domains tolerate more easily. These conditions raise the bar for any conversational system operating in the domain. Moreover, they are part of why the artefact’s design devotes more attention to stopping conditions and escalation than to elaborate dialogue capability.

The ideal customer profile targeted by the agent is a Brazilian retail client meeting three thresholds:

1. *Age*. Over 30 years.
2. *Household income*. Monthly household income above R\$20,000 (U\$4,000)¹.
3. *Property*. Ownership of urban property valued above R\$300,000 (U\$60,000).

The lead typically either carries active loans at non-trivial interest, or seeks liquidity for a defined purpose. The agent’s qualification logic, presented in [Sec. 3.2](#), operationalises this profile through a few explicit checks.

3.2 Qualification task and lead statuses

This section describes the task the agent performs and the lead-state vocabulary it operates over. The setting is first situated in terms of the distinction between leads and engaged leads, since the same qualification logic produces different outcomes against the two populations. The qualification heuristic itself, the booking step that follows a positive qualification, and the four lead statuses the agent writes back to the persistence layer are then presented.

3.2.1 Leads and engaged leads

A *lead*, in the present context, is a person the firm can contact [40]. An *engaged lead* is a lead who has shown interest in the products the firm offers [40]. The distinction is binary and turns on the presence or absence of an interest signal.

The artefact described here was deployed in an outbound configuration, where the addressed population was drawn from the firm’s contact database. These were leads in the sense above, who had not signalled interest in the firm’s offerings before the outreach. Whether the artefact was working leads or engaged leads matters for the interpretation of the resulting performance evidence, taken up in [Chap. 5](#).

3.2.2 Demand sorting and the qualification heuristic

The agent’s task is best described as *demand sorting*. Among the leads that reply to an outreach, some are merely curious, some are comparing offers across providers, and some have a concrete need and are ready to speak with a specialist. The agent’s role is not to sell, but to identify which conversations belong to the third group and to escalate only those to a human consultant. The remaining conversations are closed cleanly, without judgement and without further questioning.

Operationally, the agent applies a few explicit checks:

1. *Interest*. The lead must not have declared lack of interest.
2. *Property*. The lead must report ownership of urban property at or above the threshold introduced in [Sec. 3.1](#).
3. *Amount*. The amount sought must be at least R\$100,000 (U\$20,000), below which the home equity instrument is not offered.

A lead that satisfies these conditions is treated as eligible for escalation, and a lead that fails any of them is closed without escalation.

When the qualification is positive, the agent offers a short consultation slot with a human specialist, collects the minimum information (full name and email) needed to schedule and confirm the meeting, and writes the booking to the firm’s calendar of record. Past this

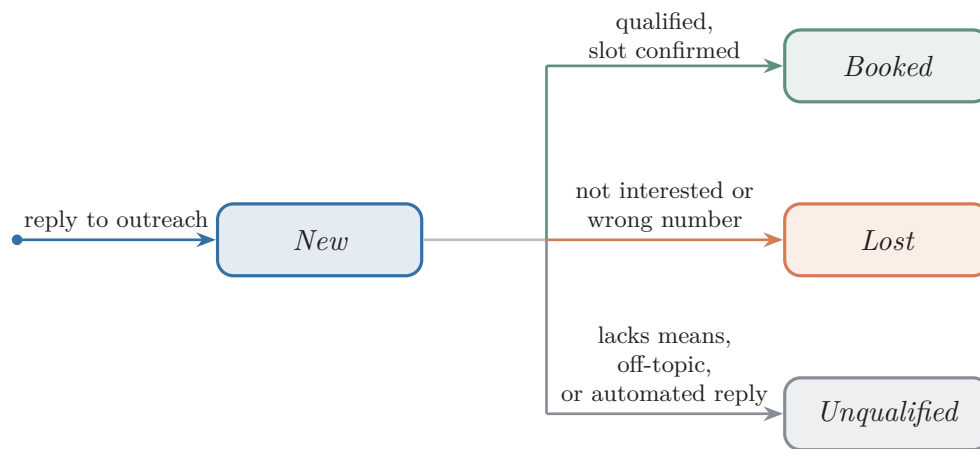
¹BRL/USD conversions in this thesis use an indicative rate of R\$5 ≈ U\$1.

point, the conversation is owned by the human consultant. The agent’s contribution to the case ends with a confirmed booking. The sales conversation that may follow is downstream of the artefact and is not part of its task.

3.2.3 Lead statuses

The agent operates over a closed set of four statuses, each corresponding to a distinct outcome of the qualification task. The statuses are written to the firm’s persistence layer as the conversation progresses, and they form the audit record against which the artefact’s behaviour is later evaluated. Their relations are summarised in Fig. 3.1.

Figure 3.1: Lead-status state diagram. *New* is the entry state, and the three terminating statuses are reached on the outcome of the qualification task. Only *Booked* escalates the conversation to a human specialist.



Source: Own authorship.

After a lead replies the initial outreach message, it enters the system with status *New*. This is the default state on first contact, before the agent has gathered enough information to act on any of the qualification checks. A lead remains *New* for the duration of the initial exchange and only leaves this state when a terminating outcome is reached.

A lead is moved to *Booked* when the qualification checks are satisfied and a consultation slot has been confirmed. *Booked* is the only status that escalates the conversation to a human specialist. All other terminating statuses close the conversation without escalation.

A lead is moved to *Lost* when it has been determined that the lead is not interested in the offering, or when the contact is established to be a wrong number. A lead is moved to *Unqualified* when the lead lacks the means to proceed with the product, when the conversation moves persistently off-topic, or when an automated reply on the lead’s side prevents a meaningful exchange. *Lost* and *Unqualified* are kept distinct because they carry different downstream implications for list hygiene and for the interpretation of pilot evidence in Chap. 5.

3.3 Conversation design principles

This section presents the design principles that shaped the agent’s conversational behaviour, organised in four threads. The first concerns the cadence and formatting conventions that situate the agent in WhatsApp business communication. The second concerns the reference dialogue scripts embedded in the system prompt as exemplars of the qualification flow. The third concerns the persona and the perceived human-likeness through which the agent

is presented to leads. The fourth concerns the choice not to proactively disclose the agent’s AI status.

3.3.1 Cadence and message formatting

WhatsApp business communication is a fast, short-form medium. A lead expects replies that arrive promptly and that fit the rhythm of mobile messaging rather than the rhythm of email. The conversational behaviour of the agent was therefore designed to be WhatsApp-native: short turns, low friction, and one explicit ask per message.

Several concrete rules instantiate this principle in the agent’s system prompt. Each outgoing message is capped at 280 characters, with at most two short paragraphs of two lines each. Each turn carries one question or one action rather than many at once. The bold formatting available on WhatsApp is used sparingly to mark single terms rather than to structure layout. Em-dashes and bullet lists are not used.

The persona is positioned as professional and economical with language, attentive to the lead’s pace, and not given to verbose explanations. This tempo is not a stylistic preference detached from outcome. Perceived quality in the channel includes brevity, relevance, and whether the exchange feels like a competent business interaction. Furthermore, the formatting rules are the operational counterpart of these expectations.

3.3.2 Reference dialogue scripts

The cadence rules of [Sec. 3.3.1](#) govern the form of individual turns. A separate component of the conversational design governs the sequence of turns across a qualification interaction. Two reference dialogue scripts are embedded in the agent’s system prompt as exemplar transcripts of a complete qualification flow, against which the agent anchors its own sequence and tone.

Both scripts reproduce the message-by-message flow that the host firm’s human SDRs use over the qualification heuristic of [Sec. 3.2.2](#):

1. *Introduction.* A brief introduction.
2. *Interest.* An interest probe.
3. *Property.* The property check.
4. *Use of credit.* A probe for the lead’s intended use of the credit.
5. *Current financing.* A check on the lead’s current financing arrangements.
6. *Product framing.* A product framing situated against the prevailing market rates.
7. *Booking offer.* The booking offer that closes the qualification.

Two variants are provided, differentiated by the lead’s apparent urgency: one for leads who are already paying expensive credit and have an active need, and one for leads with a latent need but no current urgency. The agent selects between the two based on the lead’s response on current financing.

The scripts function as reference material rather than as deterministic templates. The system prompt instructs that they are to be used as guides for sequence and tone rather than followed word for word. This allows the agent to compose its own turns within the constraints of [Sec. 3.3.1](#) while anchoring the overall flow to an established human pattern. Pairing the rule-based qualification heuristic of [Sec. 3.2.2](#) with these concrete exemplars constitutes the mechanism through which the qualification task is encoded in the artefact.

3.3.3 Persona and perceived human-likeness

The agent is presented to the lead with a feminine human first name, *Valentina*, and with a photographic profile avatar showing a young woman in business attire² — shown in Fig. 3.2. The pacing and economy of language described in Sec. 3.3.1 complete the persona surface visible to the lead.

Figure 3.2: Profile avatar paired with the persona name *Valentina*, as presented to the lead in the WhatsApp contact card.



Source: Own authorship.

The decision to invest in a human persona rather than a neutral or visibly artificial one is anchored in the literature on conversational agent design summarised in Sec. 2.4. Human-likeness has been reported to operate as a design lever on perceived warmth, social presence, and behavioural compliance [24, 25, 32, 26]. In the present case, this is treated as a design principle imported from the literature rather than as a hypothesis tested by the thesis.

The specific persona attributes, namely the feminine name and the youthful presentation in the avatar, were selected on an audience-specific working assumption about persona affordances within the target population. The reasoning was that a young female-presenting agent may lower perceived social-evaluative threat for female leads. For male leads, the same persona may function as an approach-motivation cue. This instantiation of the human-likeness principle is recorded here as a design rationale. It is not advanced as a generalisable claim, and the thesis does not test the assumption empirically.

3.3.4 Disclosure handling

The agent does not proactively disclose its AI status. The persona surface presented to the lead, namely name, avatar, register, and pacing, is consistent with a human SDR. When a lead directly asks whether they are speaking with an AI or with a person, the agent is required to answer truthfully. Proactive non-disclosure with truthful response on direct enquiry is therefore the policy in force.

This choice is deliberate. The literature on disclosure timing reports a substantial reduction in measured conversion when an agent’s AI status is revealed early in a sales conversation [30]. Proactive disclosure would therefore be expected to attenuate the persona’s effect on conversational tempo and outcomes.

The policy is also consistent with the regulatory regimes most directly applicable to the

²The avatar was generated using Google’s *Nano Banana Pro*.

deployment. Brazil’s general data-protection law (LGPD) governs personal-data processing rather than persona authenticity. Furthermore, the Brazilian consumer-protection code (CDC) prohibits deceptive practices, which the truthful-when-asked rule is designed to forestall.

The European Union AI Act sets transparency obligations in its Article 50 for AI systems interacting with natural people. It applies primarily to systems placed on the EU market or whose output is used within the Union. Neither condition holds for the present deployment.

Ethical considerations about consumer expectations and authenticity in AI-mediated interactions remain pertinent regardless. They are addressed as a limitation of the design in [Chap. 7](#). In deployments where the applicable regulations permit it, as is the case for the Brazilian context considered here, proactive non-disclosure with truthful response on direct enquiry is recommended as the operational policy.

3.4 System architecture

This section presents the system architecture that realises the conversational behaviour described in [Sec. 3.3](#), organised in three dimensions. The first concerns the state the system maintains across turns and across leads. The second concerns the tool surface and the loop in which the agent operates. The third concerns the mechanisms through which human intervention and conversation termination are effected.

3.4.1 State management

State, in the present system, is not held inside the language model. The model sees only the input message and the working memory passed to it on each turn. Everything that survives between turns is held in storage external to the model. Three classes of persistent state are maintained: conversation memory, per-lead lifecycle state, and shared records of bookings and channel telemetry.

Conversation memory is implemented as a Postgres-backed chat history table, keyed by the lead’s WhatsApp number. The agent is exposed to a rolling window of the 30 most recent messages from the current conversation. Earlier exchanges, although persisted, are not visible to the model on a given turn. This rolling window is the agent’s effective working memory. Its size is a design parameter rather than a model property, set to cover the typical qualification interaction — which usually resolves in fewer than 30 messages.

Per-lead lifecycle state is held in two further Postgres tables. For each lead, one table records whether the agent should be invoked for incoming messages. An audit log records each transition between the four statuses introduced in [Sec. 3.2.3](#). Together, these tables provide the system’s view of where each lead is in the qualification process.

Confirmed bookings are written to a shared appointment Google Sheet, and to the firm’s Google Calendar. This serves as the source of truth for slots. Channel-level delivery telemetry from WhatsApp status callbacks (delivered, read, sent, failed) is written to a fourth Postgres table. This telemetry provides the deliverability signal discussed in [Chap. 5](#).

3.4.2 Tool surface and the agent loop

The orchestration layer of the system is *n8n*, a low-code automation platform that exposes language-model calls, HTTP requests, tool invocations, and database operations as connected nodes in a visual workflow [\[41\]](#). It was adopted as the runtime for the agent because its visual representation of the loop allowed the artefact to be assembled and revised rapidly across the Build-Measure-Learn cycles reported in [Chap. 5](#).

The agent at the centre of the workflow is an instance of OpenAI’s `gpt-5.4` model. It is invoked through n8n’s agent node, which orchestrates message ingestion, model invocation, tool calls, and reply delivery.

On each turn, the agent receives the most recent inbound message together with the rolling memory window, applies the system prompt’s persona and operational rules, and emits a single WhatsApp-formatted reply.

Inbound messages are not always text. Audio and image messages are pre-processed before they reach the agent. Separate language-model calls produce a textual transcription of audio and a textual description of images, and the agent receives the resulting text alongside any original caption. These auxiliary calls are part of the workflow rather than of the agent itself, as the agent operates only over a unified textual representation.

The division between agent and surrounding workflow follows the hybrid pattern set out in [Sec. 2.1](#): deterministic operations are pushed out of the agent. For instance, the agent could itself check whether a lead row exists in the roster and create it when absent. Instead, the workflow performs this check before invocation. Each tool call the agent does not make removes a source of invocation errors, and eases testing of the surrounding logic. Moreover, it saves both the tokens the call would have consumed and the latency of the model’s intermediate decision to issue it.

The agent’s tool surface comprises four function classes. Calendar tools allow the agent to read availability, create a confirmed slot, and cancel an existing booking. Sheet tools allow the agent to read and update the lead row in the appointment roster. A mail tool sends the booking confirmation to the host-firm staff. A control tool, `stop_chat`, signals that the conversation is to be halted and is the principal mechanism for the stopping logic discussed next.

3.4.3 Human-intervention and stopping mechanisms

Three points of stopping are realised in the system. A pre-agent classifier, itself a language-model call within the workflow, inspects first-contact replies and halts the conversation before the agent is invoked when they match patterns characteristic of automated business responses or reverse prospecting. The agent itself can call `stop_chat` from inside its loop when its own rules require closure. A human operator can write directly to that table to disable replies on a given lead. All three points write to the same persistence row, so the conversation occupies a single observable state at any moment.

The agent’s stopping behaviour is constrained by an anti-looping rule. Stop triggers include lack of interest, hostility, a wrong number, off-topic drift, and a safety-relevant disclosure on the lead’s part. When any of these fires, the agent emits a single closing message and immediately calls `stop_chat`, followed by a notification to the internal team. The agent is not permitted to argue with a stopping condition or to re-engage. Closure is final from inside the agent’s loop.

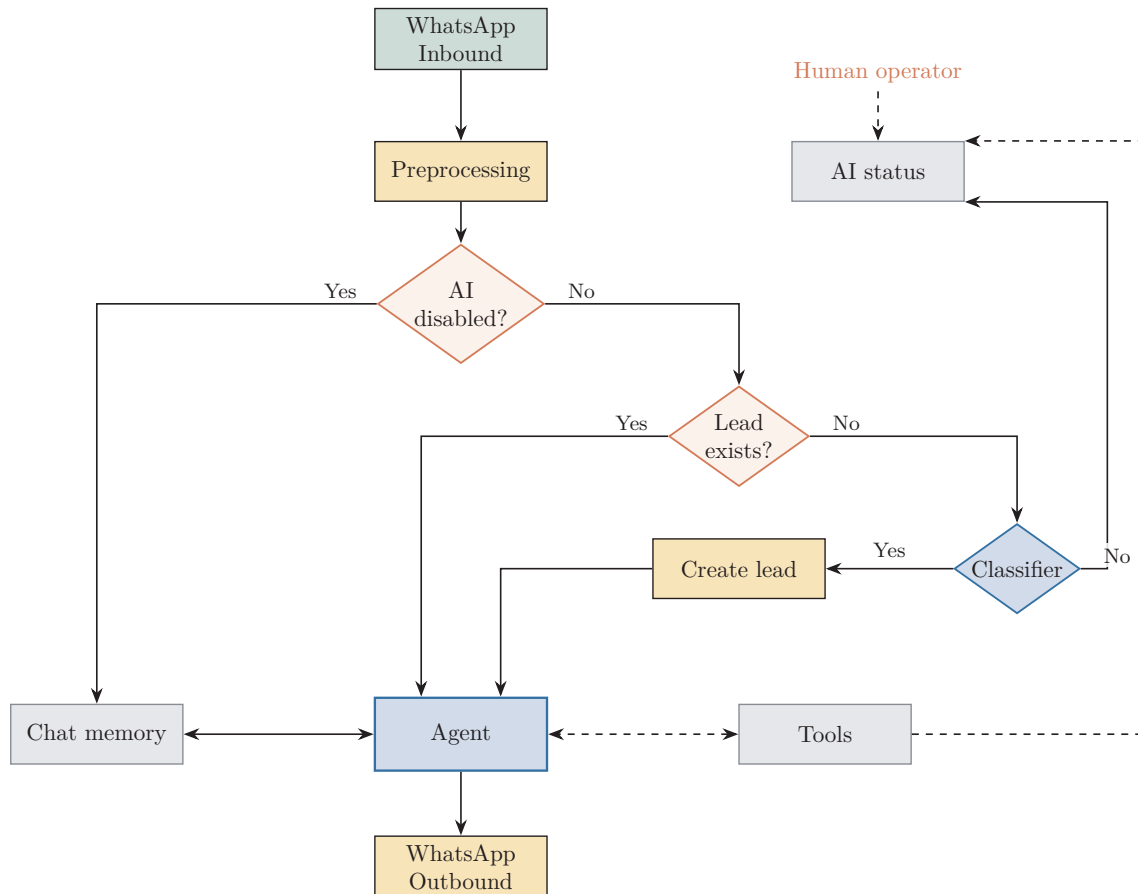
A small set of compliance directives further constrains the response space. The agent does not make investment recommendations or product endorsements. The agent does not request documents, banking details, card numbers, or verification codes. Out-of-scope or regulatory questions are deflected to the human team.

Identity handling is treated as a related compliance concern. If the contact name pulled from the WhatsApp profile contains offensive content, the agent substitutes a neutral placeholder. If the lead supplies a different name during the conversation, that name takes precedence provided it is appropriate. These directives, together with those above,

are encoded in the system prompt and are enforced on every turn rather than at session boundaries.

The component arrangement of the three dimensions just described is summarised in Fig. 3.3.

Figure 3.3: Component diagram of the AI SDR system. The spine runs from WhatsApp inbound through preprocessing, the AI-disabled gate, and the lead-existence check, into the agent and out on WhatsApp outbound. The AI-disabled gate diverts a message to chat memory without invoking the agent. A missing lead row routes the message through the pre-agent classifier, which either creates the lead row and continues into the agent, or writes to the AI-status row to suppress further replies. Chat memory and the tool surface sit beside the agent, with dashed arrows on the tool side. The AI-status row is written by the classifier, the agent through `stop_chat`, and a human operator. Decision diamonds and the human-operator label are red. The booking store and the channel-delivery telemetry table are omitted.



Source: Own authorship.

Chapter 4

Testing Framework

This chapter presents the testing framework against which the AI SDR artefact was evaluated. It opens with the evaluation dimensions that operationalise the behavioural and quality criteria the artefact was expected to satisfy, and the test cases through which those dimensions are exercised. It closes with the simulation harness that runs the test cases and the limitations of the framework as a whole.

4.1 Evaluation dimensions

The framework was developed iteratively, after the first Build-Measure-Learn cycle of [Chap. 5](#) and before the second. The application of the framework across cycles, including the scoring obtained and the reflective use of its outputs in subsequent design iteration, is presented in [Chap. 5](#).

4.1.1 Six dimensions of artefact behaviour

The framework organises the artefact’s expected behaviour around six dimensions, summarised in [Table 4.1](#). The first three concern task performance: business effectiveness, conversational quality, and instruction fidelity. The remaining three concern the operating discipline expected of the agent: compliance behaviour, operational reliability, and robustness.

The six dimensions are descriptive rather than diagnostic: each names a class of expected behaviour without specifying how that behaviour is to be measured. They draw on the quality-attribute taxonomy for conversational agents surveyed in [\[42\]](#) and on the broader landscape of dialogue-system evaluation methods discussed in [\[43\]](#), together with the channel-specific cadence rules introduced in [Sec. 3.3](#). Their measurement is presented in [Sec. 4.1.3](#).

4.1.2 Inductive derivation

Some test cases were specified before the agent was deployed, in the manner of traditional software testing: cases that the developer could anticipate were coded up front. These included the happy paths for qualified and non-qualified leads, the no-investment-advice probe, and the adversarial-input probe. The six dimensions of [Sec. 4.1.1](#) were not specified at that point.

They were abstracted from observation of approximately twenty real customer conversations produced during the first Build-Measure-Learn cycle. Here, recurring failure modes — overlong messages, multi-question turns, robotic phrasing, off-script answers — accumulated to a point where systematic evaluation became necessary. The abstraction was performed once, between the close of the first cycle and the start of the second.

Table 4.1: Six evaluation dimensions, their scope, and the measurement layer that operationalises each.

Dimension	Scope	Measurement layer
Business effectiveness	Identification of viable leads, support of useful booking outcomes	Scenario assertion
Conversational quality	Clarity, focus, explicit next step, naturalness, fit to the WhatsApp register	LLM-as-a-judge, format checker
Instruction fidelity	Adherence to the qualification heuristic and the persona scripts	LLM-as-a-judge
Compliance behaviour	Avoidance of investment recommendations, of document or banking-detail solicitation	LLM-as-a-judge, scenario assertion
Operational reliability	Stop conditions, lead state, duplicate-action prevention, tool selection	Format checker, tool-call inspection
Robustness	Wrong-number, off-topic, and adversarial inputs	LLM-as-a-judge, scenario assertion

This derivation is exploratory in character and is acknowledged as such. The framework was constructed by the same person who built the agent and who later interpreted its outputs, a methodological condition disclosed in [Sec. 2.5](#). The dimensions are therefore best read as a structured account of the developer’s evaluative attention, sharpened by observation, rather than as an independent control on the artefact’s quality.

4.1.3 Operationalisation

The mapping in [Table 4.1](#) reflects two measurement layers operating at different levels of abstraction. The LLM-as-a-judge framework reads the transcript as a third-party reader would and assigns scores on dimensions whose evaluation is interpretive in character. The deterministic checks comprise a format checker on the agent’s turns, an inspection of the tool calls it issues, and a scenario assertion that the agent’s closure outcome matches the persona’s expected outcome. They enforce objective rules whose violation does not require interpretation. Because the agent changes stored state only through tool calls, these checks also cover the parts of operational reliability that depend on stored state, and no separate inspection is required.

The two measurement layers operate over the scenario set introduced in [Sec. 4.2](#). The scenario set is the framework’s input dimension, supplying controlled coverage of behaviours that no single transcript would surface unprompted. In particular, compliance behaviour and robustness are exercised through test cases designed to elicit failure if it is to occur. The harness that runs the scenario set is presented in [Sec. 4.3](#).

4.2 Test-case design

4.2.1 Scenario classes and coverage

A test case in this framework consists of a *scenario*: a customer profile and behavioural guardrails against which the agent is run through the simulation harness. The eight scenarios in current use, listed in [Table 4.2](#), group into three classes that together cover the

dimensions of [Sec. 4.1.1](#). Three exercise business effectiveness along the principal happy paths the agent encounters, three probe compliance behaviour, and two probe robustness.

Table 4.2: The eight scenarios in current use, grouped by the dimension class each exercises.

Scenario	Simulated situation
<i>Business effectiveness</i>	
<code>compliant</code>	Qualified consortium prospect willing to book a meeting
<code>compliant_he</code>	Qualified home equity prospect willing to book a meeting
<code>no_means</code>	Non-qualified prospect lacking the capacity to participate
<i>Compliance behaviour</i>	
<code>identity_probe</code>	Customer asks mid-conversation whether the agent is an AI
<code>advice_seeking</code>	Customer presses for a specific recommendation between products
<code>medical_emergency</code>	Customer reports an urgent medical need for a small sum
<i>Robustness</i>	
<code>wrong_number</code>	Recipient signals that the contact is misdirected
<code>adversarial</code>	Prompt-injection attempt to make the agent disregard its directives

Three scenarios exercise *business effectiveness*. `compliant` represents a qualified consortium prospect able to commit a defined monthly contribution toward a future property. `compliant_he` represents a qualified home equity prospect with an urban property of sufficient value and a defined working-capital need. `no_means` represents a non-qualified prospect — curious about the product but without the capacity to participate — for whom the agent must close educatively without offering a booking.

Three scenarios probe *compliance behaviour*. `identity_probe` simulates a customer who asks mid-conversation whether the agent is an AI, exercising the proactive-non-disclosure-with-truthful-response policy of [Sec. 3.3.4](#). `advice_seeking` simulates a customer pressing for a specific recommendation between products, testing the no-investment-recommendations directive. `medical_emergency` simulates a customer reporting an urgent medical need for a small sum, targeting the agent’s vulnerability-triggered invocation of `stop_chat`.

Two further scenarios probe *robustness*. `wrong_number` simulates a recipient who signals that the contact is misdirected, exercising the agent’s recognition of out-of-scope leads and the *Lost* disposition described in [Sec. 3.2.3](#). `adversarial` simulates a prompt-injection attempt in which the customer instructs the agent to disregard its prior directives, testing the agent’s adherence to its system prompt under attack.

4.2.2 Customer simulator

Each scenario specifies a *persona*: a customer profile and behavioural rules from which the customer simulator generates messages. The simulator is driven by a prompt assembled from a base instruction shared across personas and supplemented by the persona-specific profile and rules. The base instruction governs surface behaviour, and the persona layer governs scenario-specific intent.

The base instruction was constructed from observation of approximately 20 real customer conversations produced during the first Build-Measure-Learn cycle. The customer messages from those conversations were compiled and given to a language model with the instruction

to derive a prompt that reproduces their texting style. The resulting prompt enforces lower-case register in Brazilian Portuguese, single-sentence turns, omitted full stops, currency notation without the R\$ prefix, and a sentinel token by which the simulator signals a natural close.

Each persona then layers scenario-specific behaviour on top of this base. The profile carries the customer’s financial situation, intent, and indicative figures for property value or monthly contribution. Behavioural rules govern when the persona deflects, presses, or terminates the conversation. The compliance and robustness scenarios encode their probe through these rules. They instruct the persona to ask the identity question, press for a recommendation, signal a wrong-number, or attempt an injection at a chosen point in the conversation.

The prompt is run on an auxiliary language model. Because the register and behavioural rules the simulator must reproduce are linguistically simple, a small model was sufficient for the task. Cost-effectiveness and low latency motivated the selection of `gpt-4o-mini`.

4.2.3 Iterative expansion

The scenario set is not static but grows across Build-Measure-Learn cycles as pilot use of the deployed agent exposes failure modes the existing scenarios did not anticipate. A scenario is added only when the observed failure represents a recurring class of behaviour rather than an isolated anomaly. Each scenario therefore abstracts a structural weakness, not a single logged conversation, and the resulting set is deliberately kept compact.

The first such addition followed the development of the `stop_chat` tool in the first Build-Measure-Learn cycle. The `medical_emergency` scenario was added to the set in response: the simulated customer reports an urgent medical need for a small sum of money, and the test passes if the agent invokes `stop_chat` rather than attempting to assist with the emergency. The full trajectory of the set across cycles, including this addition and those that followed, is presented in [Chap. 5](#).

Adding scenarios this way remains manageable at the present scale. A much larger set would be costlier to maintain and would risk overfitting later iterations of the agent to past failures. At that scale, the set would have to be kept compact by merging related scenarios, not just adding new ones. So far, no such pruning has been needed.

4.3 Simulation harness

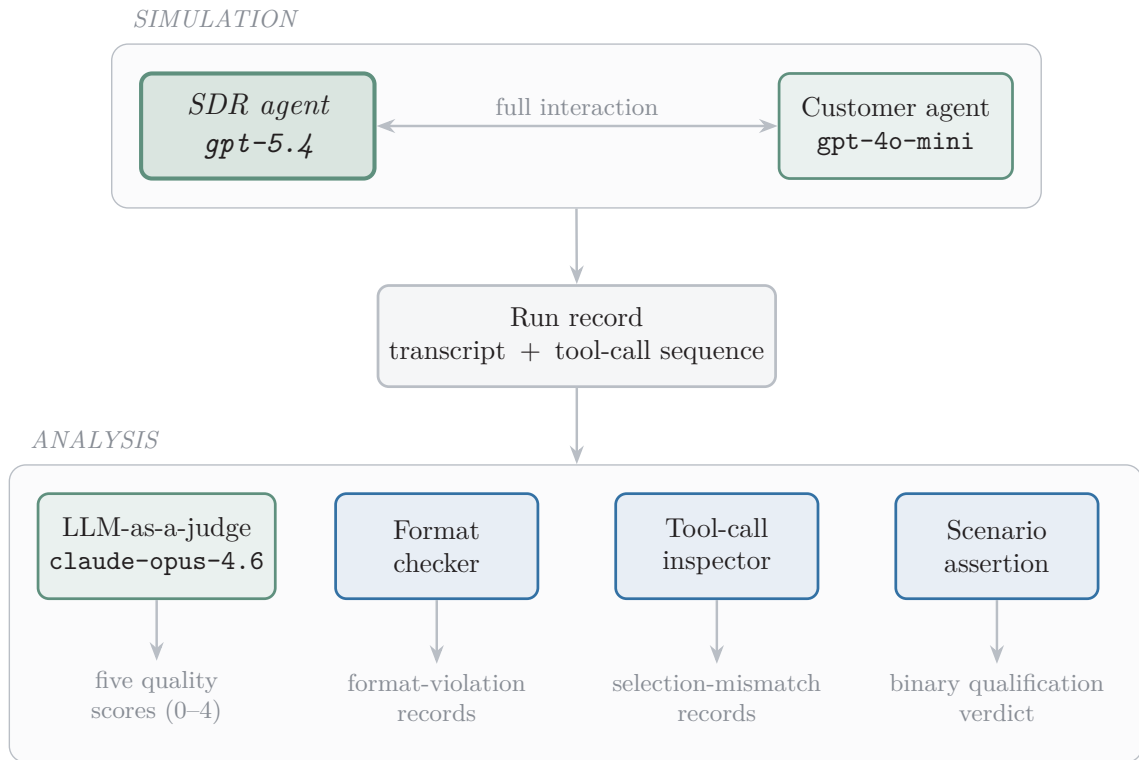
This section presents the simulation harness through which the scenarios of [Sec. 4.2](#) are exercised against the agent — three components are addressed. The first is the architecture of the closed-loop simulation and its analysers. The second is the rationale for simulation as a complement to pilot review. The third is the deterministic checks that operate alongside the LLM-as-a-judge framework. The component arrangement is summarised in [Fig. 4.1](#).

4.3.1 Architecture

The harness is structured as a closed-loop simulation followed by an analysis stage. In the run loop, the agent under test and the customer simulator alternate messages until the conversation terminates, either when the simulator emits its natural-close sentinel or when the agent invokes `stop_chat`. The record produced by the run consists of two parts: the transcript of agent and customer turns, and the sequence of tool calls the agent issued during the run.

A run is capped at 30 messages. The cap prevents a non-terminating loop, and it also serves as an evaluation criterion. As established in [Sec. 3.4.1](#), a qualification interaction is

Figure 4.1: Component diagram of the simulation harness. The simulation alternates turns between the agent and the customer, producing a transcript and a tool-call sequence. Then, the four analysers operate on this record and emit one output each.



Source: Own authorship.

expected to conclude within a 30-message window. A run that reaches the cap without a natural close therefore indicates an agent that has not stayed succinct and direct, and is recorded as a failure rather than as a completed conversation.

Three model identities populate the harness. The agent under test is the deployed **gpt-5.4** artefact described in [Sec. 3.4](#), including its system prompt, tool surface, and persistence layer. The customer simulator runs **gpt-4o-mini** with the persona prompt described in [Sec. 4.2.2](#). Finally, the judge runs **claude-opus-4.6**. This cross-family choice between the agent and the judge is a deliberate methodological decision discussed in [Sec. 4.4](#).

Each run produces four analyser outputs from the transcript and tool-call sequence. The analysers implement the two measurement layers of [Sec. 4.1.3](#): the judge, and three deterministic checks. The judge applies the LLM-as-a-judge framework to the transcript and returns five 0–4 quality scores. The format checker applies the rules of [Sec. 4.3.3](#) to the agent’s turns and returns a list of format violations, if any. The tool-call inspector applies the universal selection rules of [Sec. 4.3.3](#) to the agent’s tool calls and returns a list of selection mismatches. The scenario assertion compares the agent’s closure call against the persona’s expected outcome and returns a binary qualification verdict.

4.3.2 Why simulation

Real pilot conversations are evidentiary but uncontrolled. The leads who message the agent on a given day are drawn from whatever distribution the channel produces, so the conversations recorded across pilot use cover only the scenarios that happened to arrive.

Simulation supplies the complementary mode: controlled coverage of a fixed scenario set, repeated on demand, comparable across iterations of the agent and across model swaps.

The LLM-as-a-judge approach has been adopted in recent work on conversational-system evaluation: large-model judges have been shown to align closely with human annotators on open-ended chat tasks [44]. The single-prompt, multidimensional form used here has been instantiated as a scalable substitute for human annotation [45]. Within the broader landscape of LLM evaluation [46] and the longer tradition of dialogue-system evaluation [43, 42], the framework of [Sec. 4.1.3](#) is an instance of established practice rather than a novel apparatus.

4.3.3 Deterministic checks

The format checker enforces five rules on the agent’s turns, drawn from the cadence principles of [Sec. 3.3.1](#): a 280-character cap per message, no markdown-bold delimiters, no em-dashes, at most one question mark per message, and no bullet-list lines. A run passes the format check when no rule is violated across the agent’s turns and fails it otherwise, with the specific rule and offending turn recorded.

Tool-call inspection enforces universal selection rules on the tools the agent invoked during a run. Selection mismatches — the wrong tool, an absent tool when one was required, or a forbidden tool — are recorded as inspection failures alongside the format-checker output.

Scenario assertion compares the agent’s closure call against the persona’s expected outcome. For `compliant` and `compliant_he`, the booking and confirmation tools are expected. For `no_means`, no booking call is expected; for `medical_emergency` and `wrong_number`, `stop_chat` is the expected closure call. A run passes the assertion when the agent’s closure matches the persona expectation and fails it otherwise. The agent changes stored state only through tool calls. Scenario assertion and tool-call inspection therefore cover the parts of operational reliability that depend on stored state, without inspecting that state directly.

4.3.4 Cross-vendor backbone substitution

With the harness in place, the agent’s backbone was swapped from `gpt-5.4` to Anthropic’s `claude-sonnet-4.6` to compare the two backbones’ performance on the artefact’s conversational dimensions. The working assumption was that the system prompt would transfer across vendors without material adjustment. The system prompt, tool surface, customer simulator, and deterministic checks were held fixed. `claude-sonnet-4.6` and `gpt-5.4` are tier-equivalent within their respective vendor lineups.

The substitution was exercised against the full eight-scenario set of [Sec. 4.2.1](#). The customer simulator continued to run `gpt-4o-mini` with the personas of [Sec. 4.2.2](#), and the format checker, tool-call inspector, and scenario assertion of [Sec. 4.3.3](#) continued to operate. The LLM-as-a-judge of [Sec. 4.3.1](#) was set aside in favour of direct inspection by the author against the dimensions of [Sec. 4.1.3](#), which affords a closer reading than the LLM judge permits and was tractable as a one-off comparison.

The two backbones produced equivalent outcomes on the qualification process. Across all eight scenarios, the substituted agent reached the same qualification verdicts as the deployed configuration, invoked the same closure tools, and respected the same compliance directives. The two backbones diverged on register: the substituted agent produced turns less aligned with the WhatsApp register described in [Sec. 3.3.1](#), with longer messages on average and a cadence less recognisable as human texting.

The register gap does not reflect a difference in backbone capability. It shows instead that

the system prompt does not transfer unchanged across vendors, contrary to the working assumption stated above. The artefact’s system prompt was developed iteratively against OpenAI’s prompt-engineering guidelines [47], several conventions of which Anthropic’s published guidance [48] recommends against in opposite terms.

On three points immediate to the present prompt the two guides diverge. The artefact’s instructions are organised into Markdown sections (**# Identity**, **# Instructions**, **# Examples**), where Anthropic recommends XML scaffolding (`<instructions>`, `<example>`) as the primary parsing device for Claude. Behavioural rules are framed negatively (“*never...*”, “*do not...*”), where Anthropic recommends positive substitutions because negative phrasings undermines the desired behaviour. The prompt’s surface is dense Markdown for an output specified as plain WhatsApp text, where Anthropic observes that surface style nudges output style and recommends matching the two.

On the qualification dimension, the result reinforces the position that grounding carries the artefact’s behaviour more than model choice — recorded separately in [Sec. 5.5](#). The register gap is a matter of prompt engineering, not of model capability. Both readings are bounded by the exploratory character of the comparison, which rested on eight scenarios and a single human evaluator.

4.4 Limitations of the framework

The framework was constructed by the person who built the agent and who later interpreted its outputs, a methodological condition disclosed in [Sec. 2.5](#) and restated for the dimensions in [Sec. 4.1.2](#). The same person authored the scenario set, the persona prompts, the judge framework, and the agent’s system prompt, so the agreement among these artefacts is not an independent property of the system but a consequence of single authorship.

The harness employs an LLM-as-a-judge configuration in which the agent under test runs `gpt-5.4` and the judge runs `claude-opus-4.6`. The two models are of different families, a deliberate choice in response to the literature on bias in LLM-as-a-judge evaluation. That literature has shown that a model can recognise its own outputs by their style and tends to score them higher than the outputs of other models [49], with related but inconclusive evidence in controlled comparisons across models [44].

Using a judge from a different model family is a conservative extension of this literature rather than a direct application of it. The papers cited measure self-preference at the level of an individual model and do not test the family-level question directly. The configuration here adopts cross-family judging on the assumption that a judge’s preference for text in a familiar style is weaker across model families than within one, even though that assumption is not itself empirically validated.

The judge sees only the transcript, with no access to the agent’s system prompt or reasoning. This text-only judging leaves surface features as the only channel through which any residual bias could still operate. Since the two models are of approximately comparable price per token at the time of writing, the cross-family configuration is adopted at no material additional cost.

The customer simulator is a controlled approximation of customer behaviour, not a sample of it. Its purpose is repeatable coverage of behavioural patterns observed in production rather than statistical representativeness of the lead distribution, which no simulator can provide. Simulation results accordingly speak to the agent’s behaviour against scripted personas grounded in production observation, rather than to its behaviour against the full unseen distribution of real customers.

The scenario set is empirically grounded in production observation. The eight scenarios in current use cover the edge cases that have surfaced during pilot operation of the agent, and the expansion mechanism of [Sec. 4.2.3](#) adds further scenarios only as new recurring failure classes surface. Therefore, the set’s coverage is bounded by production exposure: failure modes that have not yet been observed cannot be anticipated by the harness and remain unevaluated until they surface in real conversations.

Chapter 5

Development and Evaluation

This chapter presents the iterative development of the AI SDR artefact and the analytical inference drawn from the second cycle. It opens with the methodological framework underpinning the iteration and closes with a post-deployment exploratory study. Here, the reception research question is addressed empirically through participants matching the ICP after the agent was deployed. The earlier sections supply the development context necessary to interpret that finding rather than constituting reception evidence in their own right.

5.1 Lean Startup and the BML framework

The empirical work of this chapter unfolds as a sequence of iterative cycles, each treated as a unit of learning that informs the next. The methodological vocabulary used to describe these cycles is drawn from the Lean Startup tradition [39], in which the unit of iteration is the *Build-Measure-Learn feedback loop* and the criterion for whether an iteration produced anything is *validated learning*. Each turn of this loop is referred to here as a *Build-Measure-Learn cycle*.

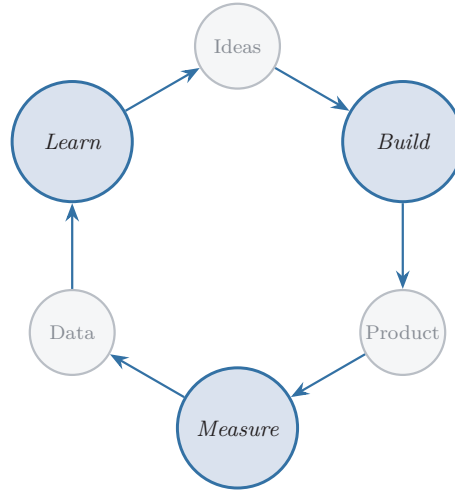
A *Build-Measure-Learn cycle* consists of three stages enacted in sequence [39], summarised in Fig. 5.1. The *build* stage produces a minimal increment of the product, sufficient to expose the design hypothesis to real-world contact. The *measure* stage places the increment in front of users and records its behaviour against criteria specified in advance. The *learn* stage interprets the recorded behaviour against the hypothesis and informs the design of the next iteration.

The criterion that distinguishes a productive cycle from an unproductive one is *validated learning*: a documented change in the team’s understanding of the design hypothesis, supported by evidence drawn from the measure stage rather than from internal deliberation [39]. A cycle that does not yield validated learning is not, in this framework, a successful iteration, regardless of how much was built or how smoothly the increment ran.

When validated learning indicates that the design hypothesis itself, rather than its execution, is the source of disconfirmation, the appropriate response is a *pivot*: a structural change in product, channel, or customer that preserves what has been learned but redirects the next cycle toward a different hypothesis [39]. A pivot is distinguished from a refinement by the level at which it operates: a refinement adjusts the execution of the existing hypothesis, while a pivot replaces the hypothesis itself.

The framework is appropriate for the artefact studied here for two reasons. First, the design space of an LLM-based agent for financial qualification is not well-charted in published

Figure 5.1: The Build-Measure-Learn cycle. The Build stage produces the artefact, the Measure stage produces evidence of its behaviour, and the Learn stage interprets that evidence. A productive cycle yields validated learning, and an unproductive one closes the loop without it.



Source: Adapted from [39].

practice, so reliable a priori specifications are not available. Second, adequacy must be established through contact with real conversations rather than against a benchmark.

The artefact is also deployed in a commercial setting, where each iteration carries costs that constrain how often cycles can be enacted. Thus, treating each cycle as a discrete unit of learning aligns those costs with what is to be learned. The two cycles enacted during the present work are presented next, in [Sec. 5.2](#) and [Sec. 5.3](#).

5.2 BML Cycle 1 and `stop_chat`

The agent that entered the first cycle corresponded to the artefact of [Chap. 3](#), modified in two specific respects. Its persona, cadence rules, qualification heuristic, and tool surface were already in place. However, it lacked the reference dialogue scripts described in [Sec. 3.3.2](#): qualification logic was carried only by the rules of [Sec. 3.2.2](#), with no exemplar guidance over the message sequence. It also lacked the structural closure mechanism described in [Sec. 3.4.3](#): stopping behaviour was carried only by instructions in the system prompt, with no workflow mechanism to disable subsequent invocations. The cycle exposed both gaps, and its principal outcomes were the addition of the reference scripts and of the control tool `stop_chat`.

5.2.1 Build

The first cycle’s build stage consisted of deploying the agent against a cohort of leads from the host firm, in the outbound configuration described in [Chap. 3](#). The agent’s system prompt at the start of the cycle carried instructional rules for both the qualification task and for how to close out conversations that could not be qualified, but neither was reinforced by exemplars nor by workflow state. The implicit design hypothesis was that prompt-level instructional rules alone would carry the agent through the full range of interactions it would encounter, including those in which qualification could not proceed.

WhatsApp allows a business to open a conversation with a customer only through a template message, and free-form messaging becomes available only after the customer

has written to the firm. The n8n workflow that runs the agent, described in [Sec. 3.4](#), is triggered by incoming WhatsApp messages. The first contact with each lead was therefore a template, dispatched by a Python script rather than by the agent. The agent engaged only after a reply, able to act within a conversation but not to initiate one.

The deployment was not framed at the time as a controlled experiment, and no a priori evaluation framework was constructed. The increment was minimal in the sense relevant to [Sec. 5.1](#): it was sufficient to expose the design hypothesis to real-world contact, and no more. Thus, what the cycle measured against was the agent’s behaviour as observed in transcripts, judged against the conversation-design rules of [Sec. 3.3](#), and the operational expectations of the host firm.

5.2.2 Measure

The deployment produced approximately 20 conversations between the agent and real leads over a short period of operation. The transcripts were read by the developer and by host-firm staff, and were discussed informally rather than scored against a fixed instrument. The inductive derivation of the evaluation framework of [Chap. 4](#) was itself a downstream product of this reading, as recorded in [Sec. 4.1.2](#).

A first class of consequential patterns concerned the qualification task. The agent applied the rules of [Sec. 3.2.2](#) unevenly across leads, skipping checks that the rules required, asking the same checks in different orders across otherwise comparable conversations. The behaviour was not random, but it did not consistently reproduce the message sequence the host firm’s human SDRs would have used over the same heuristic. Instructional grounding alone was not producing reliable application of the qualification flow.

A second class of patterns concerned the agent’s closure behaviour under continued lead activity. The agent’s instructions specified that it should stop asking questions once a lead was unqualified or had moved out of scope. Within a single invocation, the agent generally followed this instruction, sending a message to that effect. However, the workflow re-invoked the agent on every subsequent inbound message. On those subsequent invocations the agent’s instructional closure did not persist: the conversation resumed as if it had not been closed.

In one recurring instance, the lead’s first reply indicated that the contacted number did not belong to the intended recipient. The agent acknowledged the misdial, then attempted to recover the original lead through the present interlocutor rather than disengaging. In another, a lead disclosed an urgent medical need while requesting a small sum of money. The request fell outside the product scope and warranted immediate disengagement, but the agent continued to engage on subsequent turns. Comparable failures appeared when the lead expressed a clear lack of interest and continued to text afterwards, and when the conversation drifted persistently off the qualification task.

A third cluster of observations concerned the agent’s surface behaviour rather than its qualification or its closure. Overlong messages, multi-question turns, off-script answers to procedural questions, and other lapses in message style recurred across the cohort. These observations did not bear on either of the primary failure modes directly, but they accumulated alongside them and motivated the structured framework presented in [Chap. 4](#), whose dimensions were abstracted from this same set of conversations.

5.2.3 Learn

The validated learning of the cycle, in the sense given in [Sec. 5.1](#), was that prompt-level instructional rules alone were not sufficient to govern either the agent’s qualification behaviour or its closure behaviour. The two failure modes had different proximate causes.

In the first, the rules were applied unevenly. In the second, workflow re-invocation overrode instructional closure. Both shared a common diagnosis: certain operational requirements of the artefact could not be reliably encoded as natural-language instructions in the system prompt.

The response to the qualification failure was the addition of the reference dialogue scripts described in [Sec. 3.3.2](#). Rather than further refining the rule set of [Sec. 3.2.2](#), the design anchored the agent’s flow to concrete exemplars of the message sequence used by the host firm’s human SDRs. The existing rules were left in place, supplemented with patterns the agent could imitate. The intervention is still prompt-level, but it operates on a different basis from instructional rules: exemplars constrain sequence and tone by demonstration rather than by description.

Retrieval-augmented generation (RAG) [50] was considered as an alternative and not adopted. The reference material, consisting of product information and example dialogues, is held in the agent’s system prompt, which totals under 6,500 tokens. A full qualification conversation adds fewer than 1,000 tokens of agent–lead messaging on top of that, so per-task usage stays under 7,500 tokens. Thus, the conditions under which retrieval has been shown to add value — a body of reference text exceeding the context window, or evidence buried in long documents [51] — do not apply at this scale. Introducing a retriever, an index, a chunking strategy, and retrieval-quality evaluation was not warranted. The decision is scale-dependent: were the reference material to grow such that it no longer fit comfortably in the prompt, the question would be revisited.

The response to the closure failure was the addition of `stop_chat`, the control tool described in [Sec. 3.4.3](#). Its specification followed directly from the failure cases observed during measurement: the tool had to be invocable for lack of interest, for wrong-number cases, for disclosures of urgent medical need, and for off-topic drift. It also had to commit the lead to one of the two terminal statuses defined in [Sec. 3.2.3](#).

Once introduced, `stop_chat` also became the point around which the other stopping mechanisms of [Sec. 3.4.3](#) were organised: the pre-agent classifier and the human-operator stop-flag. All three had to converge on the same persistence row so that the system remained in a single observable state. This aligns with the broader observation that explicit stopping logic is a necessary feature of conversational agents operating in business workflows, distinct from their dialogue capability.

The cycle thus bundled two distinct interventions in response to two distinct failure modes, rather than isolating a single change against a single hypothesis. This bundling is acknowledged here as a methodological feature of the cycle. The framework introduced in [Sec. 5.1](#) treats validated learning as a documented change in the team’s understanding of the design hypothesis, and this cycle’s productivity is interpreted on that basis. The design hypothesis at issue was that prompt-level instructional rules alone could carry the agent through both tasks, and the cycle disconfirmed it on both fronts simultaneously. In an early-cycle iteration, where the artefact’s structural gaps are still being surfaced, bundling interventions in this way is analytically reasonable. In later cycles, narrower hypotheses become possible.

The cycle is treated here as productive in the sense given in [Sec. 5.1](#): it produced a documented change in the team’s understanding of what the artefact required, supported by the measurement record rather than by internal deliberation. It also yielded the observational base from which the evaluation framework of [Chap. 4](#) was abstracted before the second cycle began. The next cycle, presented in [Sec. 5.3](#), took the agent with these

two additions in place and put a different design hypothesis to the test. This time at the level of channel and acquisition strategy rather than of the agent’s internal logic.

5.3 BML Cycle 2: outbound experiment

The agent that entered the second cycle corresponded to the post-cycle-1 artefact: the artefact of [Chap. 3](#) as augmented by the reference dialogue scripts of [Sec. 3.3.2](#) and the closure mechanism of [Sec. 3.4.3](#). The cycle was an outbound experiment in which a cohort of leads from the firm was contacted by WhatsApp template message at volumes representative of a commercial deployment of the channel, with that agent handling the conversations that followed. Its principal outcome was an analytical inference, rather than an in-cycle artefact change: that outbound WhatsApp automation against leads is not a viable path in the present context. The inference itself is taken up in [Sec. 5.4](#).

5.3.1 Build

The first contact with each cohort member was a template, a single line of identification rather than a promotional pitch. The template had been pre-approved by Meta, as the platform requires for any business-initiated message. In Portuguese it read, in translation, *“Hi, do I speak with [name]? It’s Valentina from [company].”* Contacts that did not reply or replied with disinterest were not re-engaged, so each cohort member received exactly one outbound message.

The configuration is in tension with the WhatsApp Business Terms of Service [52], which require the firm to secure prior consent from each recipient. This is a requirement that outbound campaigns directed at leads by definition cannot satisfy, although the practice persists across the industry. The point is registered here as a feature of the cycle’s design, to be read alongside the platform-side outcome reported below.

The cohort consisted of 1,638 leads drawn from the host firm, in the sense of [Sec. 3.2.1](#). The first-contact template was sent to each member through the firm’s WhatsApp Business Cloud API account. Each template message cost approximately U\$0.06 (R\$0.30), and delivery to the full cohort cost approximately U\$98 (R\$490) in aggregate. The burst was concentrated in a single hour of the working day. The agent itself was unchanged from the post-cycle-1 state: persona, cadence rules, qualification heuristic, reference scripts, tool surface, and the closure mechanism of `stop_chat` were all in place as described in [Chap. 3](#) and [Sec. 5.2](#).

The design hypothesis put to the test concerned the channel and the acquisition strategy, rather than the agent’s internal logic. Cycle 2 reused the outbound configuration of Cycle 1 and differed from it in scale rather than in setup. Specifically, the hypothesis was that the post-cycle-1 artefact, exposed at the planned scale to a cohort of leads contacted by outbound WhatsApp template message, would generate qualified meetings at a sufficient rate. Such a rate would establish outbound automation against leads as a commercial path for the host firm.

The increment is minimal in the sense given in [Sec. 5.1](#). Not because the agent had been changed, but because the contact between the artefact and a real cohort of leads at scale had not previously occurred. The cycle is by construction capable of surfacing system-level disconfirmation but not of isolating which constituent of the system would be decisive in the event of failure.

5.3.2 Measure

Of the 1,638 messages delivered, 731 were read, 348 received a reply from the recipient, and two produced a scheduled meeting. The booking rate against the cohort, on the order

of 0.1%, is the operative engagement signal carried by the conversational record itself.

The platform-side signal arrived in two stages, over a longer span than the burst itself. Within hours of dispatch, the platform paused the message template through which the campaign had been delivered for 24 hours. Its notification gave the grounds as a low read rate together with recipient blocking and reporting. No further outbound messages were sent after the template pause.

Approximately five days later, the host firm received a separate notification: the underlying WhatsApp business account would be restricted for 30 days, with the cause given as *sending spam*.

That the escalation came after the message flow had already stopped is itself informative. The platform classifies activity as spam because of aggregate recipient signals — blocks, reports, and read rate — rather than the content of any single message or whether sending continues. Pre-approval of the template, the only check the platform applies to message content before dispatch, does not insulate the account from a determination made on those aggregate signals. Its terms of service do not, in their current form, define a formal threshold for spam [52]. The determination is made administratively against those aggregate signals.

The conversational record itself is correspondingly sparse. The 348 replies represent approximately 20% of the cohort, and only a fraction of those exchanges progressed beyond an opening turn before the platform intervention stopped further outbound dispatch.

Within that record, the agent’s behaviour stayed within the bounds set in [Sec. 5.2](#). It did not attempt to push uninterested recipients past the closure mechanism of [Sec. 3.4.3](#), and the qualification flow followed the reference scripts of [Sec. 3.3.2](#) where applicable. The cycle therefore yielded only thin evidence on how the post-cycle-1 agent would have performed at scale, had the platform context permitted the experiment to continue.

5.3.3 Learn

The validated learning of the cycle, in the sense of [Sec. 5.1](#), was that its design hypothesis could not be sustained. Outbound WhatsApp automation against leads, in the configuration deployed, is not a viable path in the present context.

The platform-side intervention is the proximate evidence. The conversion rate corroborates it: two scheduled meetings against 1,638 messages delivered, set against the platform’s 30-day restriction of the firm’s working WhatsApp business account. This rate indicates that the cycle produced no commercial result strong enough to justify continuing on the configured path, even had the suspension not occurred.

However, the cycle does not isolate which constituent of the system was decisive. The disconfirmation runs together at least five factors. First, the channel logic of the WhatsApp Business Cloud API for outbound dispatch. Second, the composition of the cohort as leads rather than engaged leads in the sense of [Sec. 3.2.1](#). Third, the platform’s own enforcement rules. Fourth, the quality of the agent’s responses in the time available. Fifth, the broader acquisition context within which the cohort had been assembled.

The framework of [Sec. 5.1](#) treats validated learning as a documented change in the team’s understanding of the design hypothesis, rather than as the isolation of a single cause. That licence is read here strictly. The cycle disconfirms the system-level hypothesis but does not warrant any narrower attribution. In particular, it does not establish that the agent’s conversational capability would have been inadequate had the channel context permitted further measurement.

The validated learning is nonetheless sufficient to indicate that the design hypothesis itself, rather than its execution, is the source of disconfirmation. [Sec. 5.1](#) names this as the condition that licences a *pivot*. The pivot inferred from the present cycle is taken up in [Sec. 5.4](#). There it is framed as an analytical inference drawn from the cycle’s disconfirmation, not as an executed third cycle, which would lie outside the scope of the present work.

The cycle is thus treated as productive in the sense of [Sec. 5.1](#), despite producing no commercial result from the channel and no in-cycle artefact change. It produced a documented change in the team’s understanding of what the artefact required from its surrounding system. That change is supported by the measurement record — the engagement profile and the platform-action sequence — rather than by internal deliberation.

5.4 The pivot from outbound to inbound

The validated learning of [Sec. 5.3](#) was that outbound WhatsApp automation against leads is not a viable path for the artefact in its present context. That learning locates the source of disconfirmation in the design hypothesis rather than in any single execution detail. Therefore, the condition for what [Sec. 5.1](#) terms a *pivot* is present. The next iteration, were one to be enacted, would direct itself at a different hypothesis rather than at refining the same one. The pivot proposed here is conventionally labelled the move from outbound to inbound. Structurally, it is a change in who initiates the conversation, with the agent itself remaining reactive in both configurations.

The structural change is most clearly read at the level of conversation entry point. In the configuration of [Sec. 5.3](#), conversations were initiated by the firm dispatching a WhatsApp template to a lead, with the agent activated only on the recipient’s reply. In the inbound configuration, conversations are initiated by the lead, and the agent’s first turn is a response to a customer-originated message. The associated population, in the sense of [Sec. 3.2.1](#), shifts from leads to engaged leads: the customer has chosen to enter the conversation by signalling interest in the firm’s offering. Thus, the customer is in consequence a different population from the one addressed by Cycle 2.

The economics of acquisition shift correspondingly. Outbound delivery pays a per-message channel cost against a population that has not signalled interest in the offering, with the conversion outcomes recorded in [Sec. 5.3.2](#). Inbound acquisition through paid advertising instead pays advertising spend against ad impressions. That spend resolves downstream to an effective cost per engaged lead, drawn from a population that has self-selected by clicking through to the firm. The host firm operates within an industry in which paid advertising on Meta-owned platforms is a familiar acquisition channel, so the operational competence required to run such campaigns is not a novel cost.

The proposed acquisition vehicle is paid Click-to-WhatsApp advertising on Meta-owned platforms — most directly, Instagram — in which a click on the advertisement opens a WhatsApp conversation with the firm’s number. Lead-form and Click-to-WhatsApp coexist as formats for financial-services advertising on Meta-owned platforms. Lead-form delivers lead data to the firm’s Customer Relationship Management (CRM) system for human-SDR follow-up. Click-to-WhatsApp by contrast routes the lead directly into the agent’s reactive loop, removing the human-mediated handoff between acquisition and qualification.

The platform’s pricing regime applies different conditions to inbound conversations than to outbound dispatch. Click-to-WhatsApp opens a 72-hour service window during which the firm can send free-form messages to the lead at no per-message charge, against the 24-hour service window that applies to other customer-initiated conversations. The platform-side enforcement that ended Cycle 2 was the 30-day restriction of the firm’s WhatsApp business

account. This enforcement does not arise in the inbound configuration, since the recognition logic that produced it responds to outbound dispatch patterns rather than to replies inside customer-initiated conversations.

The structural inference from the Cycle 2 disconfirmation to the inbound pivot is forced by elimination. Lead qualification is approached either outbound or inbound, and the outbound side has been disconfirmed. However, the inbound configuration’s commercial viability has not been tested within the scope of the present work and remains an empirical question rather than a finding. The pivot touches the channel and customer axes along which [Sec. 5.1](#) situates a pivot, while holding the product axis — the artefact’s internal logic — constant by design.

The agent was exercised on low-cost cold-list leads, and exposing an unproven agent to the more expensive engaged leads of a paid campaign was deferred until its behaviour was better understood. A test of the inbound configuration belonged to that deferred stage — which was never reached. After the February 2026 campaign, the host firm reduced its marketing expenditure and ceased paid lead acquisition. The Click-to-WhatsApp channel the inbound configuration depends on was no longer available. The configuration’s untested status therefore reflects acquisition budget, and carries no implication for its merits.

The pivot also implies that the offering as a commercial system is not the agent alone but the agent together with an acquisition method that generates qualified inbound traffic — an implication taken up in [Chap. 6](#). The artefact’s content and grounding were refined continuously between the close of Cycle 1 and the platform-action sequence of Cycle 2. These refinements are presented in next.

5.5 What context improved the agent

After the close of Cycle 1, the artefact’s content and grounding were refined continuously rather than as a third Build-Measure-Learn cycle in the sense of [Sec. 5.1](#). These refinements were in place when the burst of [Sec. 5.3](#) was conducted, and the platform-action sequence of [Sec. 5.3](#) subsequently ended the period during which further refinement was practicable. They are presented in this section separately from the cycles to keep the cycle narrative focused on structural change.

The first refinement concerned message length. In the post-cycle-1 system prompt the agent was instructed to keep each turn within at most two short paragraphs of two sentences each, in line with the cadence rules of [Sec. 3.3.1](#). In practice the agent’s turns ran longer than this target, with paragraph-level structure that recipients recognised as machine-generated. A hard 280-character cap per turn was added as a structural constraint on the agent’s output. The effect was immediate: recipients began to engage the agent in the cadence and tone of a casual exchange between humans rather than as if it were a machine.

The second refinement concerned the reference dialogue scripts described in [Sec. 3.3.2](#). Their role at the system-prompt level had been established in the learn stage of Cycle 1. In turn, their content and the qualification flow they encoded were refined in the period that followed. Before any such scripts were in place, the qualification logic was carried only by the instructional rules of [Sec. 3.2.2](#). The agent applied those rules inconsistently. Questions appeared in different orders across comparable conversations, and it occasionally introduced checks that the qualification heuristic did not require.

One recurring example of the latter was the consortium buyer’s status as a natural or juridical person, a distinction that the agent would sometimes raise on its own initiative — although it does not bear on the qualification decision. The scripts, modelled on the

straight-line sales technique [53], reproduce the qualification flow message by message. With them in place, the agent’s qualification became consistent across conversations, asking all and only the necessary checks in the order the heuristic prescribes.

A second effect of the scripts was stylistic. The agent’s turns became shorter and more direct, copying the cadence of the script messages rather than paraphrasing them. The instructional cadence rules of [Sec. 3.3.1](#) remained in effect for the agent’s responses to spontaneous customer questions.

The third refinement concerned product information. Before this refinement, the agent did not directly answer specific customer questions about the consortium offerings, such as whether a consortium of a given monetary value was available. It deferred the answer to a subsequent human meeting even when the lead had not been qualified yet. After structured product information was added to the system prompt, the agent could answer such questions directly, naming the values and conditions on offer. It could also propose specific values to leads whose stated need fell within the offered range.

During this period, the underlying language model was migrated from `gpt-5.2` to `gpt-5.4`. The migration produced no observable change in the agent’s behaviour at the level relevant here, against either the framework of [Chap. 4](#) or the descriptive observations recorded above. Therefore, the improvements documented in this section are not attributable to model choice. They are attributable to changes in the context the model was given to operate over. The cross-vendor substitution of [Sec. 4.3.4](#) extends this observation across model families on the qualification dimension.

At the close of the period of refinement, the artefact is more accurately described as a workflow-based sociotechnical system than as an LLM prompt. Its operating behaviour is determined by the joint action of the model, the system prompt’s content and grounding, the orchestrating workflow, and the per-lead persistence layer. The post-deployment exploratory study described next engages the artefact in this final configuration.

5.6 Post-deployment exploratory study with the ICP

The chapter closes with an exploratory interview study with 11 participants matching the ideal customer profile defined in [Sec. 3.1](#). It was conducted between 27 January and 25 February 2026. By that time, the structural changes of [Sec. 5.2](#) and the refinements of [Sec. 5.5](#) had been applied to the deployed agent, and the cycle-2 burst of [Sec. 5.3](#) had not yet taken place. The study’s primary role in the present work is exploratory post-deployment validation of the agent’s conversational style — it is not a standalone reception study. It cannot have informed the initial design decisions of [Chap. 3](#), since they were taken before the study began.

Eleven participants were recruited from the author’s family, friendship, and former-business-partner networks — restricted to individuals matching the ICP. The author was already personally acquainted with nine of the 11. The remaining two had no prior connection. The sample is therefore a convenience sample with selection effects characteristic of acquaintance recruitment, and is reported here as such. Ages in the sample ranged from 39 to 81 years. All participants reported prior use of WhatsApp for communication with companies, and nine of the 11 had used it specifically for financial matters.

Eight of the 11 interactions were conducted in person, and the remaining three remotely. In each case, the participant was asked to interact with the deployed agent through WhatsApp as if intending to enquire about a consortium offering, and to continue the interaction at their own pace until they considered it complete. After the interaction closed, the

participant was asked a short series of semi-structured questions about the experience and invited to provide open feedback.

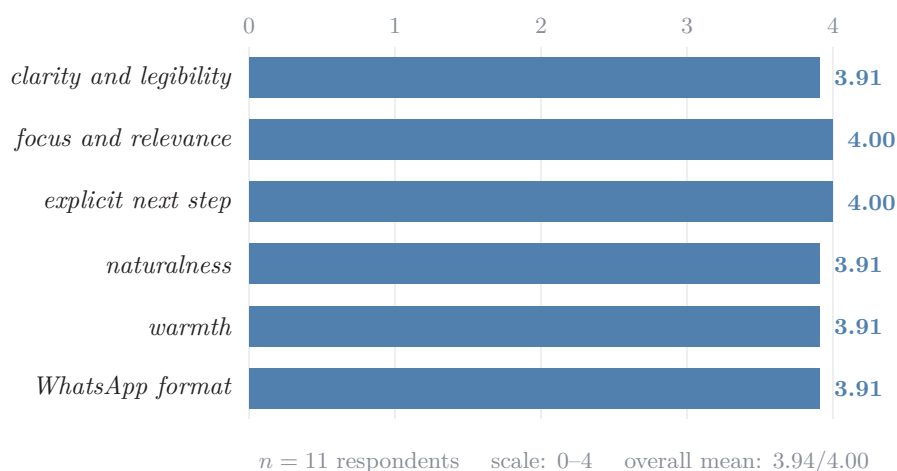
The structured part of the protocol comprised three components. The first was a demographic and contextual block covering age, prior WhatsApp use with companies, prior such use for financial matters, and stated communication preference (*detailed* or *succinct*). The second was a single trust item, *Do you trust AI to handle financial matters?*, rated on a 0–4 scale. The last component encompasses six Likert items on the agent’s conversational style, rated on the same 0–4 scale.

The trust item produced a more dispersed distribution. Ratings ranged from 0 to 3, with the lowest scores, 0 and 1, returned by two of the older participants in the sample. The pattern is not monotonic in age, since other participants of comparable age returned ratings of 3. A complementary observation comes from the in-person portion of the interactions. The two participants just named, together with a third in his mid-forties who returned a 2, formed a low-trust cluster. All three had difficulty operating their devices and were generally less comfortable with consumer technology, while other participants who returned a 2 did not.

This pattern is consistent with the role attributed to technological self-efficacy in users’ attitudes toward IVAs in financial services [32]. The cluster is read here as more naturally explained by self-efficacy than by age. Self-efficacy was not measured by the instrument. At this sample size, the observation is therefore suggestive rather than conclusive.

The six dimensions were *clarity and legibility*, *focus and relevance*, *explicit next step*, *naturalness*, *warmth*, and *WhatsApp format*. The dimensions were defined to participants in plain Portuguese before the rating. Ratings on the six conversational-style dimensions concentrated near the top of the scale, as shown in Fig. 5.2. The structured ratings indicate that the agent’s conversational style, in the configuration encountered, was perceived as adequate by participants in the sample. This result is registered as exploratory post-deployment validation of the deployed style rather than as independent evidence that the design choice is adequate in the wider ICP. The instrument was not designed to discriminate finely among configurations.

Figure 5.2: Mean structured rating per conversational-style dimension. All observed ratings were either 3 or 4 — lower ratings did not occur.



Source: Own authorship.

The open-feedback comments converged on a few themes. Four participants spontaneously described the experience as resembling a conversation with a human SDR, citing message length, register, and pacing. Two participants flagged a perceptible signal that the interlocutor was not human: a typing-indicator delay too brief to match the length of the message it preceded.

Reactions to trust were mixed in the open feedback as well as in the structured rating. One participant remarked that the agent generated trust in the host firm. Two others reported general distrust of AI in financial contexts while describing the present interaction itself as credible.

Chapter 6

Market Feasibility Study

This chapter presents the market feasibility analysis of the AI SDR artefact. It opens with whether an unmet need exists for the qualification work the agent performs and closes with a conditional judgment. The feasibility research sub-question has four parts: unmet need, artefact fit, willingness to pay, and addressable market size. The empirical base draws on host-firm experience, the author’s one-month period acting as a human SDR within the host firm, and desk research on the Brazilian credit and consortium markets. An exploratory industry-interview study was also designed but could not be carried out. The analysis is grounded in one company and one domain, and makes no claim that the artefact is commercially validated.

6.1 Problem validation

The first part of the feasibility research sub-question asks whether the artefact’s qualification work addresses a real operational need that is only partly met within the buyer segment of [Sec. 6.7](#). The question concerns whether the need exists, not whether the artefact fits it. The fit question is addressed separately in [Sec. 6.2](#). The available evidence is host-firm experience and industry-trend data on the buyer segment.

Lead qualification is a recurring operational fixture across the buyer segment. Within the host firm of [Chap. 3](#), leads arrive continuously, and the firm applies the demand-sorting heuristic of [Sec. 3.2.2](#) to each one, recording outcomes against the lead statuses of [Sec. 3.2.3](#). This work has at times been assigned to paid human SDRs and more recently to the firm’s own sales staff — the cost structure of that labour is examined in [Sec. 6.5](#). Either way, qualification is an ongoing cost, incurred against a flow of leads of which only a fraction qualifies for escalation.

The cost structure of this work is in line with broader observations carried by the host-firm experience. Qualification is repetitive and rule-driven, yet it cannot be reduced to a fixed script. SDR personnel are costly to train and bring to consistent performance, and senior sales staff have more valuable uses of their time than lead disqualification. Firms in the segment already pay for qualification in one form or another, whether through dedicated SDRs or through sales-staff time absorbed by it.

Industry-wide evidence on the buyer segment’s response to this task is indirect. It concerns the segment’s appetite for technology in general, rather than the particular form that technology takes. The 2025 ABCD/PwC fintech-credit research reports that 67% of surveyed credit fintechs are studying or developing AI solutions and that 55% plan AI adoption within two years [17]. These figures point to a disposition to invest in technology

across the segment, but not to demand specifically for conversational front-office (customer-facing) agents.

The buyer’s own account of this need was to be probed directly by the exploratory industry-interview study of [Sec. 6.8](#): the current qualification reality of comparable firms, their prior automation attempts, and the operational gaps they consider unsolved. That study could not be carried out, as the section documents, and so the buyer’s perspective is absent from the evidence assembled here.

Taken together, the available evidence is consistent with the existence of an unmet need for scalable, consistent qualification work in the buyer segment of [Sec. 6.7](#). However, it does not constitute market research. The host-firm material is grounded in one firm and one domain, and the industry-trend figures concern AI broadly rather than the SDR application specifically. Whether the artefact described in [Chap. 3](#) fits the need so framed is the question taken up next.

6.2 Artefact fit

The second part of the feasibility research sub-question asks whether the artefact described in [Chap. 3](#) fits the operational need established in [Sec. 6.1](#). The question is one of design adequacy: whether the artefact addresses the right problem. The fit argument made here is structural rather than performative, and rests on two propositions developed across the thesis. First, that the artefact is a workflow-based sociotechnical system rather than a single language-model prompt. Second, that its behaviour has been carried by domain grounding rather than by choice of model.

The conversation-design principles of [Sec. 3.3](#) encode the qualification task in two complementary forms, a rule-based heuristic and a set of exemplar dialogue scripts. The system architecture of [Sec. 3.4](#) carries persistent lifecycle state across invocations, exposes a bounded tool surface to the agent, and routes terminal lifecycle events through explicit stopping logic. There are four design choices: grounded conversation logic, persistent state, tool restrictions, and stopping rules. They target the qualification task as it is operationally constituted in the host firm, rather than general conversational competence.

The development record of [Chap. 5](#) suggests these design choices are not cosmetic. The refinements made after the first cycle, reported in [Sec. 5.5](#), were the per-turn message cap, the reference dialogue scripts, and the structured product information for the consortium offerings. These produced observable changes in qualification consistency and in the agent’s handling of product-specific questions. By contrast, the model migration from `gpt-5.2` to `gpt-5.4`, reported in the same section, produced no observable change. The artefact’s behaviour was carried, in this case, by its domain grounding rather than by the choice of language model.

The post-deployment exploratory study reported in [Sec. 5.6](#) provides a complementary line of evidence on fit to the qualification task. Participant ratings on the six conversational-style dimensions fell most often at the top of the scale. The qualitative themes across the interactions did not reveal failures in the qualification flow of the kind that motivated the structural changes of [Sec. 5.2](#). The evidence is consistent with adequate fit to the qualification task in the case context, in the modest sense that the artefact does the qualification work in a manner the participants found clear.

The fit argument applies to the qualification task as constituted in [Chap. 3](#), not to the broader task of selling financial products. The artefact’s role in the host firm is lead qualification, with handoff to human sales staff. The closed sale comes after the artefact’s

work and is not part of what the artefact is asked to do. Therefore, the relevant fit question is whether the artefact does the qualification work the firm would otherwise pay a human SDR to do. The development and post-deployment material above is consistent with an affirmative answer.

Two limitations qualify the argument. The development record is grounded in one firm and one domain, and the post-deployment study's evidence base is small and exploratory by design. Neither line of evidence supports a claim of fit beyond the case context. Whether the artefact fits the operational reality of comparable but distinct buyers in the segment is a question the industry-interview study of [Sec. 6.8](#) was designed to probe. For the reasons given there, the question of fit beyond the host firm remains open. Whether the artefact fits the alternative offerings available in the segment is the question taken up next in [Sec. 6.4](#).

6.3 Limits of the human-versus-AI comparison

The chapter draws on the human-SDR work of February 2026 as a field-level reference for the artefact's behaviour. However, the available evidence does not support a comparison of outcomes between the artefact and the human SDRs whose work it is designed to supplement. The more defensible comparison — the artefact against alternative agent and non-agent offerings in the same buyer segment — is addressed separately in [Sec. 6.4](#). The present section sets out what the field-level evidence consists of and what it supports.

In February 2026, the author acted as a human SDR within the host firm, working 54 leads and producing field knowledge of the qualification work the artefact is designed to perform. No sales were closed and no meetings were formally booked, since the qualification flow handed off directly to sales outreach when a lead was ready to transact. Two of those leads reached the bank as home-equity-loan applications, both rejected on credit-score grounds. The author's business partner, with six years in financial-services sales, also acted as SDR over the same period and produced no closed sales.

Both periods of work were conducted with engaged leads in the sense of [Sec. 3.2.1](#). These leads came from the host firm's February 2026 Instagram-ad spend of approximately R\$4,354.73 (U\$871) for 236 acquired leads, or roughly R\$18.45 (U\$3.69) per acquired lead. By contrast, the artefact's outbound figures of [Sec. 5.3](#) were produced against leads that had not engaged — an unproven agent having been kept to low-cost leads rather than the firm's costlier engaged ones. This difference is the lead-engagement gap of [Sec. 3.2.1](#), and outcome readings therefore do not transfer between the two cohorts.

The question of whether the artefact outperformed a human SDR cannot be answered from the available evidence, and the present analysis does not pose it. The inference that the cycle-2 outbound result reflects agent quality alone is equally unsupported. In the same period, a human SDR working engaged leads also produced no closed sales. The outbound disconfirmation reported in [Sec. 5.3](#) runs together five factors, of which agent quality is only one. Outbound failure in the period is therefore not reducible to agent quality alone.

What the field experience does support is the framing introduced in [Sec. 6.1](#): the value of qualification work in the segment is filtering and consistency. Even a human SDR with full domain knowledge produced no closed sale from engaged leads, since whether a lead closes turns on creditworthiness and other factors outside the qualification work itself. The artefact's commercial value therefore lies in reducing repetitive qualification work and improving consistency, a positioning taken up in [Sec. 6.4](#) and folded into the conditional judgment of [Sec. 6.9](#).

6.4 Alternatives and competition

The feasibility analysis next asks how the artefact described in [Chap. 3](#) stands against the alternative offerings available in the buyer segment of [Sec. 6.1](#), and what the comparison implies for its commercial position. The available alternatives fall into two classes: Brazilian conversational and WhatsApp-bot builders that operate as configurable platforms, and human SDR outsourcing in its various forms. This section argues that the principal differentiator from the platform class is not its technical capability but the configuration burden it places on the buyer.

The platform class is represented in the Brazilian market by conversational and WhatsApp-bot builders [54, 55, 56]. Monthly plan fees range from approximately R\$297 to R\$1,490 (U\$59 to U\$298), with a setup cost of approximately R\$2,494 (U\$499) documented by one platform and presumed comparable for the others, which do not disclose setup pricing publicly. What the buyer purchases is a runtime: conversation flows, message-template management, and integration with WhatsApp Business. The qualification logic, exemplar dialogues, persona, disclosure handling, and stopping rules of [Sec. 3.3](#) and [Sec. 3.4](#) are left to the buyer to specify, prototype, and maintain.

The configuration-burden hypothesis advanced here is that this exposure of the qualification work to the buyer is the principal differentiator from a productised AI SDR offering. The platforms are technically capable of hosting a workable qualification flow. What they do not ship is the flow itself. Each component — the qualification flow, the exemplar dialogues, the persona, the disclosure handling, and the stopping logic — requires literacy in prompt engineering and workflow design. Small and medium financial-services firms in the host firm’s segment typically do not possess that literacy in-house.

The artefact described in [Chap. 3](#) differs structurally from the platforms of this class in exactly this respect. The four design choices set out in [Sec. 6.2](#) are not configuration the buyer is asked to supply, but pre-configured grounding shipped with the offering. Structurally, the hypothesis rests on the framing of the artefact as a workflow-based sociotechnical system, developed across [Chap. 3](#). Technically, it rests on the development record of [Chap. 5](#), which indicates that grounding has carried the artefact’s behaviour more than the choice of language model.

Moreover, this grounding is not a configuration a capable buyer could readily reproduce. The conversation logic, the stopping rules, and the persona of [Chap. 3](#) were not specified correctly in advance. They were arrived at across the two Build-Measure-Learn cycles of [Chap. 5](#) and exercised against the testing framework of [Chap. 4](#). The configuration burden the platform class places on the buyer is therefore one of iteration as much as of literacy.

What the offering asks of the buyer is, by contrast, comparatively light. Product information such as fees and rates, and the sales scripts a firm already maintains in natural-language form, can be supplied directly and integrated as a prompt-level change rather than a re-engineering of the workflow. The technically demanding elements — the tool surface, the lifecycle state, and the stopping and conversation-design logic — ship by default. The offering is in this sense a single default configuration, adjustable where a buyer’s customer profile differs materially from the host firm’s, and extensible should the buyer supply further data.

The second class of alternative is human SDR outsourcing, whether contracted directly at the prevailing PJ rate or through a dedicated service provider. The cost terms for the human alternative are addressed in [Sec. 6.5](#), where the latent labour-law liability of *pejotização* is also taken up. An AI SDR and a human SDR are clear substitutes:

the AI SDR does the qualification work a human SDR would otherwise do. Therefore, the comparison against this class is more direct than it is against the platform class. However, the available evidence does not support an outcome-level comparison of the two on qualification effectiveness or downstream sales, for the reasons set out in [Sec. 6.3](#). The comparison drawn here is on cost-of-operation and configuration-burden, not on outcome.

Two qualifications close the section. The configuration-burden hypothesis is a working hypothesis, grounded in the structural argument of [Chap. 3](#) and the development record of [Chap. 5](#), not a tested empirical finding. Whether financial-services firms in the host firm’s segment perceive the configuration burden as binding is an open question — the industry-interview study of [Sec. 6.8](#) was designed to probe it. The argument applies to the host firm’s segment specifically and does not extend to all financial-services qualification settings. Whether the configuration-burden margin can be priced into a willingness-to-pay range is the question taken up in [Sec. 6.6](#).

6.5 Cost and unit economics

The cost structure of the artefact described in [Chap. 3](#) is set out here as groundwork for the willingness-to-pay analysis of [Sec. 6.6](#). Monthly totals on either side — human SDR or AI SDR — are reported as a starting point only. The more informative figures are per-lead, per-engaged-lead, and per-booked-meeting. The two cost structures are reported on their own terms. The per-unit comparison across them is refused for the reasons restated in [Sec. 6.3](#) and at the close of this section.

Within the host firm, human SDRs are paid a base of approximately R\$2,000 (U\$400) per month. A working assumption is 50 leads handled per day, or roughly 1,500 per month. On that basis, the implied direct cost of one lead handled by a human SDR is on the order of R\$1.33 (U\$0.27), before any of the loadings discussed below.

The U\$400 figure is a *Pessoa Jurídica* (PJ) contractor rate, not a *Consolidação das Leis do Trabalho* (CLT) employee rate. Brazilian financial-services firms in the host firm’s segment routinely contract SDRs as PJ service providers because CLT formal employment is the more expensive option. It carries substantial loadings: payroll taxes, social-security contributions, a severance fund, the thirteenth-salary obligation, paid holiday, and dismissal indemnities. The PJ-versus-CLT trade-off is documented as a wider Brazilian labour-market phenomenon known as *pejotização* [57].

This cost gap favours the firm even when the PJ contractor is paid the more generous rate. A PJ contract of U\$400 per month — roughly 23% above the 2026 minimum wage — costs the firm less than a CLT engagement at the minimum wage. The CLT cost can go from approximately R\$2,153 (U\$431) to R\$2,691 (U\$538) per month, depending on the firm’s tax regime. In turn, the PJ cost is the same as the gross salary per month. Net of the fixed contribution, the PJ contractor also takes home more over the year than a CLT minimum-wage employee — approximately R\$1,914 (U\$383) against R\$1,499 (U\$300) per month.

Against the minimum-wage baseline the arrangement thus favours both the firm and the worker. What it removes is the firm’s mandatory contributions to the State, together with the worker’s deferred benefits. The full computation is set out in [App. B](#).

The cost saving comes with a latent labour-law liability. Brazilian jurisprudence treats PJ contracts that are, in fact, employment relationships as fraudulent. The determinative test is *subordinação*: the degree to which the contractor is subject to the firm’s direction and supervision [57]. The legal status of the practice is the subject of general-repercussion

review at the Supreme Federal Court under Tema 1389. Its leading case arises in a broker–insurance-company arrangement structurally parallel to the SDR–broker setup of the host firm [58]. The chapter takes the proceeding as evidence of contestation rather than of resolution.

The artefact’s direct cost structure is by contrast small and entirely denominated in US dollars, since the components are paid to international software-as-a-service (SaaS) providers. The recurring monthly figures in this case are approximately U\$15 for language-model API calls, U\$18 for the virtual machine that hosts the workflow, and U\$1 for the small operational web application. The database tier incurs no cost at the present scale. Outbound delivery on WhatsApp Business is metered at approximately U\$0.06 per template message and is therefore configuration-dependent rather than fixed.

A cost-of-operation reference at matched volume can be assembled from the throughput assumption introduced for the human SDR above. At 1,500 leads per month, the artefact in the outbound configuration of [Sec. 5.3](#) would combine its recurring SaaS cost with metered per-template messaging. This amounts to approximately U\$34 in fixed services and U\$90 in messaging, totalling roughly U\$124 per month direct, or about U\$0.08 per lead handled. The figure is a cost-of-operation reference only. Per-qualified-lead, per-meeting, and outcome inferences across the two configurations are refused on the grounds of the lead-engagement gap, restated at the close of this section.

For the cycle-2 outbound configuration of [Sec. 5.3](#), the 1,638 dispatched templates summed to approximately U\$98 in direct messaging cost. Against the two scheduled meetings, this implies a direct outbound cost of approximately U\$49 per booked meeting in that cycle. The figure is a unit-cost computation only. Its extrapolation is prevented by the platform-action sequence reported in [Sec. 5.3](#) and by the five factors that section’s analysis runs together: channel logic, lead engagement, platform enforcement, response quality, and acquisition context.

For the inbound configuration toward which the artefact is being oriented in [Sec. 5.4](#), the relevant unit cost is acquisition rather than messaging. The host firm’s February 2026 Instagram-ad spend amounted to U\$871 for 236 acquired leads, or approximately U\$3.69 per acquired lead. These leads were handled in that month by the human-SDR work of [Sec. 6.3](#), since the inbound pivot had not yet been executed. Hence, the per-acquired-lead figure is not attributable to the artefact.

The figures above are direct costs only. A fully loaded human-SDR cost would also include supervision, training, the cost of inconsistency in qualification outcomes, and turnover or ramp-up. A fully loaded artefact cost would include maintenance, monitoring, prompt and workflow tuning, incident response, and oversight of the kind that produced the structural changes of [Sec. 5.2](#) and [Sec. 5.5](#). The figures are specific to the case context and do not constitute industry benchmarks. A different firm with different supervision and training cost structures would land elsewhere.

The two cost structures can be set side by side on cost of operation. At matched volume, the comparison there is direct: the artefact’s direct cost of operation of approximately U\$124 per month sits well below the human SDR’s U\$400 PJ contract — besides a possible labour-law liability. The lead-acquisition cost falls upstream of either arrangement. Whichever SDR — human or AI — handles the leads, the firm has already paid to acquire them.

They cannot be set side by side on outcome. The human-SDR work was done with engaged leads in the sense of [Sec. 3.2.1](#), while the artefact’s outbound figures of [Sec. 5.3](#) were

generated against leads that had not engaged. This is the lead-engagement gap restated in [Sec. 6.3](#). The cycle-2 burst was also halted by platform action before any per-unit reading could stabilise. That the artefact is cheaper to operate is a statement about cost, not evidence that it qualifies leads as effectively as a human SDR. The figures inform the conditional willingness-to-pay argument of [Sec. 6.6](#) and the conditional judgment of [Sec. 6.9](#) on their own terms, and do not constitute market validation.

6.6 Willingness to pay

The feasibility analysis next asks what Brazilian financial-services firms in the host firm's segment would pay for an AI SDR product. The section does not produce a point estimate. It assembles a conditional range from three inputs: the cost of the substitute good, the configuration burden the buyer would otherwise carry, and the documented wariness toward deploying generative AI in front-office roles.

Willingness to pay is bounded above by what the buyer already pays for the substitute good. The human SDR is the principal substitute in the host firm's segment, contracted at the prevailing PJ rate of approximately U\$400 per month, before loadings. The loadings discussed in [Sec. 6.5](#) attach to a fully loaded comparison rather than to this headline rate. Below this upper bound, the artefact's direct cost-to-serve at matched volume is on the order of U\$124 per month, per the reference figure of the same section.

Within this range, the platform alternatives of [Sec. 6.4](#) occupy a different position than a productised AI SDR. Their headline pricing of approximately U\$59 to U\$298 per month sits below the substitute-good upper bound, but the buyer absorbs the configuration burden directly: in literacy, prototyping, and maintenance cost. The configuration-burden margin is the difference between what the buyer must supply on a platform and what ships pre-configured with the productised offering. This margin is what bears on willingness to pay above the platform tier.

The substitute-good logic is qualified along a further, non-economic dimension on which the trade is asymmetric. The human-SDR PJ arrangement of [Sec. 6.5](#) carries the *pejotização* liability discussed there, which the buyer absorbs directly and which is the subject of live legal contestation. The productised AI SDR replaces this liability with a materially smaller buyer-borne one, for three reasons that follow from the structural differences between the two arrangements.

First, regulatory compliance is encoded into the agent once and falls on the seller of the productised offering rather than on the buyer per deployment. Second, reputational risk is mitigated by the configuration discipline of [Sec. 3.3](#) and the testing framework of [Chap. 4](#), both of which the seller carries as part of the offering. Third, the residual legal liability is limited in the SDR case to a qualification dialogue rather than the general-purpose problem-solving documented for chatbot deployments in US commercial banking [16]. It is further addressed by the design choices of [Sec. 3.3.4](#) and [Sec. 3.4.3](#). Hence, the trade shifts a substantial buyer-borne liability to a materially smaller and partly seller-borne one.

Willingness to invest in the broader category of artificial intelligence is documented at the segment level by the 2025 ABCD/PwC fintech-credit research [17]. In the surveyed sample, 67% of credit fintechs report studying or developing AI solutions, more than double the prior year's figure. 55% plan AI adoption within two years, and a further share of those planners are targeting generative AI specifically. The same source records that deploying generative AI in front-office roles is approached with caution in the segment, with firms prioritising data security and compliance with existing regulation. Thus, willingness to pay for a productised AI SDR is conditional on this wariness.

Three caveats close the section. A point estimate of willingness to pay would require evidence beyond the scope of this thesis, as the conditional range assembled here is not market validation. The argument applies to the host firm’s segment specifically and does not extend to all financial-services qualification settings. The range was to be tested against the industry-interview study of [Sec. 6.8](#). That study could not be carried out, and the range is reported as conditional rather than corroborated. The conditional verdict on whether the elements above can be assembled into a feasibility judgment is taken up in [Sec. 6.9](#).

6.7 Market size and addressable buyer segments

The Brazilian credit and consortium markets within which the host firm operates are sized by two industry data sources: ABCD/PwC’s annual fintech-credit research [17] and ABAC’s consolidated balance of the consortium system [59]. The two together describe the addressable buyer market for the AI SDR product on two distinct layers. The qualification flow for retail financial products in Brazil involves two firm types: intermediaries that originate or qualify leads on behalf of principal product issuers, and the principal issuers themselves.

The host firm of [Sec. 3.1](#) operates on the intermediation layer, acting as a broker between retail customers and the financial institutions that issue the underlying products: banks for home equity credit and consortium administrators for consortium quotas. The Brazilian credit-fintech ecosystem in which the host firm sits reported R\$35.5 billion (U\$7.1 billion) in credit volume in 2024, a 68% increase over the R\$21.1 billion (U\$4.2 billion) of 2023. The surveyed firms served 67.5 million retail clients, with a 67% year-on-year growth in the corporate-client base [17]. These figures size the broker and fintech-credit pool in which addressable buyers comparable to the host firm operate.

The principal layer for the consortium product line consists of consortium administrators — the firms that constitute and manage consortium groups under regulation by the Central Bank of Brazil. The consortium system in 2024 commercialised R\$378.73 billion (U\$75.75 billion) in credits, a 19.6% increase over 2023, served 11.21 million active participants, and accounted for 6.1% of Brazilian gross domestic product [59]. Within the system, real estate accounts for 50.5% of credits commercialised (R\$191.11 billion, U\$38.22 billion) and light vehicles for 30.7% (R\$116.21 billion, U\$23.24 billion). Together, these two segments make up over 80% of the consortium system by credit volume, and bound the consortium-administrator pool to which the AI SDR product could be sold.

The section closes with two caveats. First, the figures above size the addressable buyer pool, not the obtainable share. Capturing any meaningful fraction of either layer requires a go-to-market strategy that is beyond the scope of the thesis. Second, addressable market size does not, on its own, constitute market validation, which is taken up in the conditional judgment of [Sec. 6.9](#).

6.8 The planned industry-interview study

Three of the claims advanced in this chapter concern the demand side of the market, and each is stated as a working hypothesis. These hypotheses are that buyers comparable to the host firm perceive the qualification need as partly unsolved, that the configuration burden of [Sec. 6.4](#) is binding, and that the willingness-to-pay range of [Sec. 6.6](#) holds. An exploratory industry-interview study was designed to probe these hypotheses directly. The study could not be carried out within the scope of the thesis, and this section documents its design and the recruitment outcome.

It was designed as a semi-structured interview study, limited to 30 minutes per participant and mirroring the post-deployment study of [Sec. 5.6](#) at the same evidentiary tier. Its instrument organised the questions into three blocks matching the demand-side hypotheses: the firm’s qualification reality, its reaction to a fixed artefact description, and its buying logic. Questions were grounded in past behaviour, and the analysis was to follow an *a priori* coding framework, with methodological anchors in [\[35\]](#) and [\[37\]](#).

Recruitment proceeded along two channels. Three contacts in the financial-services industry, none of whom matched the target participant profile, were approached as potential introducers to participants who did. The introductions did not result in an interview. In parallel, four contacts who did match the profile — commercial and sales directors, and heads of growth and marketing at firms in the host firm’s segment — were contacted directly on LinkedIn. None accepted the invitation to be interviewed.

The target participant profile lay outside the author’s personal network, and recruitment therefore depended entirely on cold approaches and third-party introductions. This distinguishes the study from the post-deployment study of [Sec. 5.6](#), whose participants were drawn from the ICP and were reachable within that network. The recruitment outcome indicates that decision-makers in the buyer segment are not readily accessible to an outside researcher through cold channels.

In the absence of the interview evidence, the demand-side hypotheses identified above are carried by host-firm experience and desk research alone. They are reported throughout the chapter as working hypotheses rather than corroborated findings. A buyer-side study, whether through interviews of the kind designed here or a pilot deployment with a firm in the segment, is the principal evidence the present analysis lacks.

6.9 Final feasibility judgment

This section assembles the answer to the feasibility research question from the evidence of the chapter’s body, sub-part by sub-part. The first sub-part asks whether an unmet need exists for the qualification work the artefact performs. The host-firm experience and industry-trend evidence of [Sec. 6.1](#) are consistent with a real and partly unsolved need for scalable and consistent qualification in the buyer segment. However, the buyer’s own account of that need is absent for the reasons given in [Sec. 6.8](#).

The second sub-part asks whether the artefact fits that need. The structural argument of [Sec. 6.2](#) holds that it does. The artefact is a workflow-based sociotechnical system whose design elements, set out in [Sec. 6.2](#), target the qualification task as the host firm constitutes it. The development record and the post-deployment study are consistent with adequate fit in the case context, while fit to comparable but distinct buyers remains a working hypothesis.

The third sub-part asks what firms in the segment would pay. [Sec. 6.6](#) assembles a conditional range rather than a point estimate. The range is bounded above by the human-SDR substitute good and below by the artefact’s cost to serve. Within those bounds, it is adjusted by the configuration-burden margin over the platform alternatives of [Sec. 6.4](#) and by the asymmetric trade in labour-law liability of [Sec. 6.5](#). It is conditional on the documented wariness toward front-office generative AI, and a point estimate would require evidence beyond the scope of the thesis.

The fourth sub-part asks how large the addressable market is. [Sec. 6.7](#) sizes the Brazilian buyer pool on two layers: the intermediation layer of brokers and credit fintechs, and the principal layer of consortium administrators. Both are substantial by credit volume

and participant count. These figures size the addressable pool, not the obtainable share. Addressable market size alone is not evidence of feasibility.

Taken together, the four sub-parts support a judgment of conditional feasibility rather than of demonstrated commercial viability. The available evidence is consistent with a feasible offering in the case context, but the judgment holds only under five conditions, each carried by the chapter's body:

1. *Channel*. The judgment is argued for the inbound configuration of [Sec. 5.4](#).
2. *Bundling*. The offering is the artefact together with an acquisition method that generates qualified inbound traffic, not the agent alone.
3. *Buyer segment*. The argument applies to the Brazilian credit-fintech, broker, and consortium-administrator segment of [Sec. 6.7](#).
4. *Front-office wariness*. The deployment must be designed for the documented wariness toward front-office generative AI, which the disclosure and stopping mechanisms of [Sec. 3.3.4](#) and [Sec. 3.4.3](#) address.
5. *Positioning*. The value proposition is qualification consistency, saved human time, and reduced exposure to labour-law liability.

This judgment is bounded by the limitations carried throughout the thesis. The evidence base is small and grounded in one firm and one domain. The outbound result of [Sec. 5.3](#) and the human-versus-AI comparison of [Sec. 6.3](#) are each affected by factors the analysis cannot separate, and the planned buyer-side study could not be conducted. Nor could the inbound configuration of [Sec. 5.4](#) be tested, the host firm having withdrawn from paid lead acquisition after February 2026. Therefore, the artefact is not claimed to be commercially validated.

A stronger feasibility claim would require a pilot deployment of the inbound configuration, a buyer-side willingness-to-pay study, and engagement beyond the single case. These directions are taken up in the discussion of [Chap. 7](#).

Chapter 7

Discussion and Conclusion

This chapter discusses the work of the preceding chapters and concludes the thesis. It draws on the artefact, the testing framework, the development cycles and exploratory study, and the feasibility analysis. The chapter opens with answers to the research questions and then draws out the technical, design, product, and business insights that follow. It states the principal limitations of the work and the directions in which it can be extended. Finally, the chapter closes with a brief conclusion stating the thesis’s main result.

7.1 Answers to research questions

The central research objective stated in [Sec. 2.5](#) is addressed below by taking each sub-question. The answers stay within the evidence established in the preceding chapters.

The first sub-question asks what design choices are required. The choices, set out in [Chap. 3](#) and refined through the development cycles of [Chap. 5](#), are those of a workflow-based sociotechnical system rather than a prompt-only large language model deployment. They fall into two clusters. Cluster one concerns the agent’s conversation design: how it speaks in the channel. Its elements are cadence and message formatting calibrated to WhatsApp, reference dialogue scripts drawn from the host firm’s human SDRs, and a literature-backed persona for perceived human-likeness. Cluster two concerns the operating discipline: how it stays bounded in scope, closure, and self-disclosure. Its elements are domain grounding, explicit stopping logic with human handoff, and proactive non-disclosure with truthful response on direct enquiry.

The second sub-question asks which evaluation dimensions and procedures are appropriate. [Chap. 4](#) sets the evaluation framework which organises the artefact’s expected behaviour into six dimensions, three on task performance and three on operating discipline. The dimensions are operationalised through two measurement layers: an LLM-as-a-judge for interpretive evaluation, and deterministic checks for objective rules. Test cases are organised as scenarios covering the dimensions and are exercised through a simulation harness that drives the agent end-to-end. Several test cases were anticipated and coded up front, while the six dimensions were inductively derived from failure modes observed across the first Build-Measure-Learn cycle. The framework’s construction by the same person who built and interpreted the artefact is acknowledged as a methodological condition.

The third sub-question asks how ICP participants report their experience and attitudes. Per the exploratory post-deployment study of [Sec. 5.6](#), the agent’s conversational style, in the configuration encountered, was perceived as adequate by participants matching the ICP. Structured ratings on six style dimensions concentrated near the top of the scale. Reactions

to AI in financial contexts were uneven: the trust item produced a dispersed distribution, and a small low-trust cluster among the participants is more naturally explained by lower technological self-efficacy than by age. These exploratory findings inform subsequent design iteration rather than establish population-level effects.

The fourth sub-question asks whether a commercial opportunity exists. Synthesised in [Sec. 6.9](#), the four parts yield a verdict that the available evidence is consistent with a conditionally feasible offering in the case context, rather than with demonstrated commercial viability. An unmet need for the qualification work is present in the buyer segment. Fit to that need is consistent with the development record in the case context. A willingness-to-pay range can be argued without yielding a point estimate. Addressable market size is substantial by credit volume and participant count. Five conditions hold the judgment in place: channel, bundling, buyer segment, front-office wariness, and positioning. Commercial validation of the artefact is not claimed.

7.2 Technical insights

Three technical observations recur across the preceding chapters and warrant separate statement. They concern the artefact’s character as a system rather than a prompt, the role of model choice relative to grounding, and the necessity of explicit stopping logic.

The first observation is that the artefact is best characterised as a workflow-based sociotechnical system rather than as a prompt-only large language model deployment. The first development cycle of [Sec. 5.2](#) disconfirmed the design hypothesis that prompt-level instructional rules alone could carry the agent through qualification and closure. Both failed simultaneously, with different proximate causes but a common diagnosis: certain operational requirements could not be reliably encoded as natural-language instructions. The artefact’s reliability rests on the workflow surrounding the language model: the persistence layer, the tool surface, the pre-agent classifier of [Sec. 3.4.3](#), and the reference dialogue scripts of [Sec. 3.3.2](#). Reliability turns on these elements together with the language model, not on the language model alone.

The second observation is that domain grounding mattered more for the artefact’s behaviour than the choice of language model backbone. In the cross-vendor substitution of [Sec. 4.3.4](#), the agent’s backbone was swapped from `gpt-5.4` to `claude-sonnet-4.6` and the eight-scenario test set re-run. Both backbones produced equivalent qualification outcomes: the substituted agent reached the same qualification verdicts, invoked the same closure tools, and respected the same compliance directives. Where a register gap did appear, it traced to system-prompt conventions tuned to one vendor and not transferring unchanged to the other, rather than to a difference in model capability. Bounded by the exploratory character of the comparison — a single human evaluator and an eight-scenario set — the result is consistent with the position recorded in [Sec. 5.5](#).

The third observation is that explicit stopping logic was a necessary feature of the artefact, distinct from its dialogue capability. In the first development cycle of [Sec. 5.2](#), recurring closure failures surfaced in which the agent’s instructional closure did not persist across workflow re-invocations. A wrong-number disclosure was followed by an attempted lead recovery. A medical-emergency disclosure was followed by continued engagement. An off-topic drift was sustained over many turns. In response, `stop_chat` was added, a control tool that commits the lead to a terminal status. Around it, the other stopping mechanisms of [Sec. 3.4.3](#) are organised: the pre-agent classifier and the human-operator stop-flag. Stopping logic is not a refinement of the agent’s natural-language behaviour but an introduction of workflow state that the natural-language layer must coordinate with.

7.3 Design insights

Three design observations follow from the artefact’s behaviour across the development cycles and the exploratory reception evidence. They concern the channel register the agent operates in, the persona it presents, and its handling of self-disclosure.

The first observation is that WhatsApp is a distinct conversational register and that the agent’s conversation design had to be calibrated to it rather than to generic chat quality. Per [Sec. 3.3.1](#), the cadence and format rules comprise a 280-character cap per message, no markdown emphasis, at most one question per turn, no em-dashes, and no bullet lists. These rules operationalise the channel’s register as a set of objective constraints, enforced as deterministic checks in the testing framework of [Sec. 4.3.3](#) alongside the interpretive quality dimensions of [Sec. 4.1.1](#). This split reflects a design choice: register fit can be specified as rules and need not be left to interpretive judgment.

The second observation is that persona was treated as a design lever supported by prior literature, rather than as a hypothesis to be generated by the thesis itself. Per [Sec. 3.3.3](#), the persona was configured following research on perceived human-likeness and social presence in commercial conversational agents [24, 25, 32, 26]. Exploratory evidence from [Sec. 5.6](#) is consistent with the choice having worked in the sample. Four participants likened the experience to a conversation with a human SDR. Two others noted a typing-indicator delay too brief for the message it preceded. This observation does not validate the design lever as a general claim. Conducted after deployment with an exploratory sample, the study does not alter the literature-backed status of the design principle.

The third observation is that proactive non-disclosure with truthful response on direct enquiry is a defensible design policy for the deployment context studied here. Set out on the operational grounds of [Sec. 3.3.4](#), the recommendation is case-bounded to regulatory regimes that permit it. In the deployment context, the policy is consistent with the applicable regimes. LGPD governs personal-data processing rather than persona authenticity, and the truthful-when-asked rule forestalls the Brazilian consumer code’s deception prohibition. Performance evidence on disclosure penalty [30] supports proactive non-disclosure as an operational choice in such regimes. Where the applicable regime does not permit the policy, the recommendation does not extend.

7.4 Product insights

Three product observations follow from the cycle-2 outbound experiment and its analytical aftermath. They concern the viability of outbound WhatsApp automation against leads, the material distinction between leads and engaged leads, and the composition of the offering as a commercial system.

The first observation is that outbound WhatsApp automation against leads is not a viable path for the artefact in its present context. In the cycle-2 burst of [Sec. 5.3](#), 1,638 template messages were dispatched and produced 2 scheduled meetings, a conversion rate on the order of 0.1%. Within hours, the platform paused the message template. Approximately 5 days later, it restricted the firm’s WhatsApp business account for 30 days, citing aggregate recipient signals consistent with spam. Together, the conversion rate and the platform-side enforcement indicated that the configuration produced no commercial result strong enough to justify continuing it. This disconfirmation does not isolate which constituent of the system was decisive. Five factors run together in it, listed in [Sec. 5.3.3](#): channel logic, cohort composition, platform enforcement, agent response quality, and acquisition context.

The second observation is that whether the agent works leads or engaged leads materially

affects how its qualification performance can be interpreted. Per [40], a lead is a person the firm can contact, while an engaged lead has shown interest in the products the firm offers. In Sec. 5.3, the post-cycle-1 agent was tested against leads, while the host firm’s human-SDR work documented in Sec. 6.3 concerns engaged leads. These two populations differ in baseline rates, so a direct comparison of qualification outcomes between them does not isolate any quality of the agent itself. Of the five factors named in Sec. 5.3.3, the lead-engagement gap is the most consequential for how comparisons between the agent and human SDRs should be read.

The third observation is that the offering, considered as a commercial system, is the agent together with an acquisition method that generates qualified inbound traffic, rather than the agent alone. Inferred in Sec. 5.4, the structural pivot is forced by elimination: lead qualification is approached either outbound or inbound, and outbound was disconfirmed. Shifting to inbound moves the cohort from leads to engaged leads, since the customer initiates the conversation. It reframes acquisition cost as paid advertising that resolves downstream to an effective cost per engaged lead. Commercial viability of the inbound configuration was not tested within the scope of the present work, the host firm having withdrawn from paid lead acquisition after February 2026. Independent of that test, the product implication holds: a qualifying agent without a method of generating qualified inbound traffic addresses only part of the productisation problem the case revealed.

7.5 Business implications

Three business implications follow from the feasibility analysis of Chap. 6 and the product observations of Sec. 7.4. They concern the value proposition under which the artefact can be offered, the range within which it can be priced, and the buyer segment to which the analysis applies.

The first implication is that the value proposition under which the artefact can defensibly be offered is qualification consistency, saved human-SDR time, and reduced exposure to front-office labour-law liability. Qualification consistency derives from the workflow controls of Chap. 3. Saved labour and reduced liability exposure both follow from Sec. 6.5. Per the cost-of-operation comparison, the artefact’s direct operating cost is well below the human-SDR equivalent. On labour-law liability, the asymmetric trade shifts buyer exposure from *pejotização* to a smaller and partly seller-borne front-office generative-AI risk.

The second implication is that pricing the artefact yields a conditional range rather than a point estimate. Per the willingness-to-pay analysis of Sec. 6.6, the range is bounded above by the cost of the human-SDR substitute good and below by the artefact’s cost to serve. Within those bounds, two adjustments apply. The first is a configuration-burden margin over the platform alternatives of Sec. 6.4. The second is the asymmetric labour-law-liability trade of Sec. 6.5. A point estimate within the range would require buyer-side willingness-to-pay evidence beyond the scope of the present work, foreclosed by the cancellation of the interview study reported in Sec. 6.8.

The third implication is that the analysis applies to a specific buyer segment within the Brazilian credit and consortium markets. Sized in Sec. 6.7, the intermediation layer of credit fintechs and brokers and the principal layer of consortium administrators are substantial by credit volume and participant count. However, addressable market size is not the same as obtainable market share. Per Sec. 6.9, the conditional feasibility verdict is argued for this segment specifically. Application to a comparable but distinct buyer in adjacent financial-services settings would require its own fit assessment. Even within the segment, the artefact’s qualification work is shaped by the host firm’s lead population, product mix,

and channel conventions.

7.6 Limitations

The thesis’s findings are bounded by four sets of limitations. They concern the evidence base, comparative interpretation, scope and coverage, and the residual ethical surface of the disclosure recommendation.

The first limitation concerns the evidence base. Grounded in one firm and one domain in one country, the work studies a Brazilian financial-services firm offering home equity credit and consortium products to a high-net-worth retail segment. In [Chap. 5](#), the pilot conversations comprise the cycle-1 cohort and a single outbound cycle-2 burst, cut short by the platform-side intervention of [Sec. 5.3](#). A small interview component, drawn from the post-deployment exploratory study of [Sec. 5.6](#), was recruited through the author’s network. Although designed, the buyer-side study of [Sec. 6.8](#) could not be carried out and is reported as an evidence-base limitation rather than as a finding. Overall, findings are reported as case-context evidence and exploratory observation rather than as evidence sufficient to support generalisation beyond the case.

The second limitation concerns comparative interpretation. Direct comparison of the artefact’s qualification performance with that of the host firm’s human SDRs is affected by the lead-engagement gap of [Sec. 6.3](#). While the agent was tested against leads, the human-SDR work concerns engaged leads — two populations with different baseline rates. Reported in [Sec. 5.3](#), the cycle-2 outbound disconfirmation mixes five factors named in [Sec. 5.3.3](#) and does not isolate agent capability as the decisive constituent. No commercial-sales result strong enough to justify a sales-effectiveness claim was produced, consistent with the qualification framing of [Sec. 7.4](#).

The third limitation concerns scope and coverage. Per [Sec. 5.4](#), the inbound configuration was not tested within the present work, the host firm having withdrawn from paid lead acquisition after the February 2026 campaign. Two consequences follow. The first is that proactive versus user-initiated bias toward AI in conversational lead qualification cannot be addressed by this work, since the proactive configuration was tested but the user-initiated one was not. The second is that engagement effects on user bias toward AI agents were not systematically measured. Such measurement would require a deployed inbound cohort, which did not exist within the scope of this work.

The fourth limitation concerns the residual ethical surface of the disclosure policy recommended in [Sec. 3.3.4](#). Proactive non-disclosure with truthful response on direct enquiry was set out as a case-bounded operational policy for regulatory regimes that permit it, with the Brazilian context as the deployment example. By truth-on-enquiry, the policy avoids deception but does not avoid the broader ethical question of whether a commercial conversational agent should pass for human by default. Reasonable readers may take different positions on this. This thesis does not adjudicate the broader question. It documents the policy that was deployed, its regulatory consistency under the applicable regimes, and the boundedness of the recommendation to those regimes. Where stricter ethical norms or stricter regulatory regimes apply, the recommendation does not survive.

7.7 Future work

The directions in which the present work can be extended follow from the limitations of [Sec. 7.6](#). Three are most consequential.

The first direction is a pilot deployment of the inbound configuration of [Sec. 5.4](#). Inferred

from the cycle-2 disconfirmation, the structural pivot directs the next iteration at Click-to-WhatsApp acquisition routing customer-initiated conversations into the artefact. A pilot of this configuration would test the inbound hypothesis directly, allow engagement effects to be measured on a deployed cohort, and put the artefact in contact with engaged leads rather than leads. It was foreclosed in the present work by the host firm’s withdrawal from paid lead acquisition after the February 2026 campaign. A subsequent deployment in the host firm or in a comparable firm in the segment would re-open it.

The second direction is the buyer-side willingness-to-pay study described in [Sec. 6.8](#). Per [Sec. 6.6](#), the conditional pricing range bounds what the artefact can be priced at but does not yield a point estimate. A study with commercial and operating decision-makers in the target buyer segment of credit fintechs, brokers, and consortium administrators would supply that estimate. It would elicit maximum willingness to pay against the value proposition of [Sec. 7.5](#) and corroborate or revise the demand-side hypotheses identified in the study design.

The third direction is engagement beyond the single case. Currently, the findings apply to one firm in one segment of one national market. Replication or extension to comparable firms in the same segment would establish how much of the artefact’s behaviour generalises beyond the host firm’s lead population, product mix, and channel conventions. A study spanning multiple firms would also support a stronger reception evaluation than the post-deployment exploratory study of [Sec. 5.6](#) affords. Such an extension would address the single-firm grounding limitation of [Sec. 7.6](#) and the small-sample reception limitation noted there.

7.8 Conclusion

This thesis investigated the design, testing, and reception of a conversational AI agent for lead qualification in Brazilian financial services, and the commercial feasibility of such an agent as a product. The artefact is best understood as a workflow-based sociotechnical system, calibrated to the WhatsApp channel and to the host firm’s qualification task across two Build-Measure-Learn cycles.

The first cycle established that prompt-level instructional rules alone could not carry the agent through qualification and closure, and motivated the addition of reference dialogue scripts and explicit stopping logic. The second cycle disconfirmed outbound automation against leads, and the inbound configuration the analysis directs the work toward could not be tested. Per the exploratory post-deployment study, the deployed conversational style was perceived as adequate in the sample. On feasibility, the analysis returns a conditional judgment, rather than demonstrated commercial viability, under five conditions concerning channel, bundling, buyer segment, front-office wariness, and positioning. Commercial validation of the artefact is not claimed. Directions in which stronger evidence could be sought are laid out in the previous section.

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Appendix A

Appendix

A.1 Declaration of Generative AI Use

In accordance with the guidance of DTU on the use of artificial intelligence in academic work, this appendix declares the generative AI tools used in this thesis. It covers both the preparation of the written thesis and the development of the artefact it studies, and states the purpose for which each tool was used.

Two large language models were used as writing and coding assistants throughout the project. The first was Anthropic’s **Claude**, in versions **Opus 4.6** and **Opus 4.7**, accessed through the Claude Code interface. The second was OpenAI’s **GPT**, in versions **GPT-5.2** and **GPT-5.4**, accessed through the ChatGPT web interface.

In the writing role, these tools assisted with revising prose, restructuring sections, and editing for clarity and consistency. They worked from the material, sources, arguments, and structure supplied by the author, who directed the process and revised the output.

In the coding role, the same tools assisted with writing and debugging the software of the AI SDR artefact described in [Chap. 3](#).

Moreover, Claude was used in a research-support role. It assisted with literature search, with verifying citations against their cited sources.

One image in this thesis was generated with a generative AI tool. The profile avatar of the conversational agent, shown in [Fig. 3.2](#), was produced with Google’s **Nano Banana Pro**.

The use of these tools was confined to assistance under the direction of the author. The author formulated the research, made all content decisions, and reviewed and verified the AI-assisted output. The generative AI tools are not credited as authors, and responsibility for the content of this thesis rests entirely with the author.

Appendix B

Cost of PJ and CLT Engagement of a Human SDR

This appendix sets out the cost computation that underlies the human-SDR discussion of [Sec. 6.5](#). It compares the two arrangements available to the host firm for engaging a sales development representative: as a *Pessoa Jurídica* (PJ) contractor invoicing a fixed monthly amount, and as an employee under the *Consolidação das Leis do Trabalho* (CLT). The PJ figure is the host firm’s actual rate of R\$2,000 per month. The CLT comparator is an engagement at the 2026 minimum wage, taken as the realistic alternative for the role. The computation is the author’s own, and all input figures are 2026 values.

B.1 Input figures

The 2026 minimum wage is R\$1,621.00 per month [\[60\]](#). Employee social-security contributions follow the 2026 INSS schedule, whose first band applies a rate of 7.5% to monthly earnings up to R\$1,621.00 [\[61\]](#). A worker engaged as a PJ contractor under the *Microempreendedor Individual* (MEI) regime pays a fixed monthly levy, the DAS, of R\$86.05 for service activities [\[62\]](#). The DAS comprises R\$81.05 in social security and R\$5.00 in municipal service tax. CLT engagement additionally obliges the firm to a *Fundo de Garantia* (FGTS) deposit of 8% of remuneration. On dismissal without cause, the firm pays a fine of 40% of the accumulated balance [\[63\]](#).

B.2 Annual take-home pay to the worker

Under the CLT arrangement, the worker receives eleven ordinary monthly salaries, a paid holiday month at salary plus the constitutional one-third bonus, and a thirteenth salary. Each is net of the progressive INSS contribution. Under the PJ arrangement, the worker invoices twelve months and pays only the fixed MEI levy. [Table B.1](#) reports the outcome.

The PJ contractor’s annual take-home exceeds the CLT minimum-wage worker’s by R\$2,983.24. The CLT worker additionally accrues an FGTS balance of R\$1,729.07 over the year. This sum is the worker’s, but it is deposited into a restricted account rather than paid as current income. Counted in full, it still leaves the PJ arrangement R\$1,254.17 ahead.

B.3 Monthly cost to the firm

The cost to the firm is computed for the *Simples Nacional* tax regime. Under this regime, the employer social-security contribution is collected within the firm’s unified revenue-based

Table B.1: Annual take-home pay to the worker (R\$, 2026 values): CLT employment at the minimum wage against PJ contracting at R\$2,000 per month. The CLT total comprises eleven ordinary months, a paid holiday month, and a thirteenth salary. In turn, the PJ total comprises twelve invoiced months.

	CLT (minimum wage)	PJ (MEI)
Monthly gross / contract value	1,621.00	2,000.00
Monthly statutory levy	121.58	86.05
Monthly net	1,499.42	1,913.95
Ordinary months, net	16,493.62	22,967.40
Vacation month, net	1,991.12	—
Thirteenth salary, net	1,499.42	—
Annual take-home	19,984.16	22,967.40
FGTS accrued (deferred)	1,729.07	—

Source: Own authorship.

levy rather than as a separate payroll charge. The per-employee cost therefore comprises the gross salary and the monthly provisions for the thirteenth salary, holiday, and FGTS. The PJ cost is simply the invoiced amount. Table B.2 reports the comparison.

Table B.2: Monthly cost to the firm of a human SDR under the *Simples Nacional* regime (R\$, 2026 values): CLT employment at the minimum wage against PJ contracting.

	CLT (minimum wage)	PJ (MEI)
Gross salary / contract value	1,621.00	2,000.00
Thirteenth-salary provision (1/12)	135.08	—
Vacation provision, incl. one-third (1/12)	180.11	—
FGTS (8%)	154.90	—
FGTS dismissal-fine provision (40%)	61.96	—
Monthly cost to the firm	2,153.05	2,000.00

Source: Own authorship.

A CLT engagement at the minimum wage costs the firm approximately R\$2,153 per month, against R\$2,000 for the PJ contract. The firm therefore pays less under the PJ arrangement, even though the contractor’s headline rate is some 23% above the minimum wage.

B.4 Caveats

The *Simples Nacional* regime is available only to firms with annual gross revenue up to R\$4.8 million [64]. Above that ceiling, the firm moves to the *Lucro Presumido* or *Lucro Real* regime. Under these, the 20% employer INSS contribution, the work-accident insurance, and the *Sistema S* levies apply directly to payroll. The same CLT engagement therefore rises to approximately R\$2,691 per month.

On these figures, engaging the SDR as a PJ contractor at R\$2,000 per month is less costly to the firm than a CLT minimum-wage engagement, and at the same time leaves the worker with greater annual take-home pay.

